

1. What is cache memory?

Cache is very fast and small memory that is placed in between the CPU and the main memory. Cache memory is used to reduce the average memory access times. This is done by storing the data that is frequently accessed in main memory addresses therefore allowing the CPU to access the data faster. This is due to the fact that cache memory can be read a lot faster than main memory. *Cache memory* is an intermediate form of storage between the registers (located inside the processor and directly accessed by the CPU) and the RAM.

The functions of cache memory are:

- The CPU uses the cache memory to store instructions and data that are repeatedly required during the execution of programs, thus improving the performance and speed of the whole system.
- It also avoids the need to access the dynamic RAM to retrieve the same data repeatedly.

2. what is Byte?

The byte is a unit of digital information that most commonly consists of eight bits, representing a binary number. Historically, the byte was the number of bits used to encode a single character of text in a computer and for this reason it is the smallest addressable unit of memory in many computer architectures.

Q3: what is the function of recognition of speed of window?

Windows Speech Recognition (WSR) is a speech recognition component developed by Microsoft for the Windows Vista operating system that enables the use of voice commands to control the desktop user interface; dictate text in electronic documents and email; navigate websites; perform keyboard shortcuts; and to operate the mouse cursor. It also supports the creation of custom macros to perform additional tasks. **Functions**

WSR allows a user to control a computer, including the operating system desktop user interface, through voice commands. Applications, including most of those bundled with Windows, can also be controlled through voice commands. By using speech recognition, users can dictate text within documents, email, and forms; control the operating system user interface; perform keyboard shortcuts; and move the mouse cursor.

WSR uses a local speech profile to store information about a user's voice. Accuracy of speech recognition increases through use, which helps the feature adapt to a user's grammar, speech patterns, vocabulary, and word usage. Speech recognition also includes a tutorial to improve accuracy, and can optionally review a user's personal documents—including email—to improve its command and dictation accuracy.

Q4: What is file?

A file is an object on a computer that stores data, information, settings, or commands used with a computer program. In a GUI (graphical user interface), such as Microsoft Windows, files display as icons that relate to the program that opens the file. For example, the picture is an icon associated with Adobe Acrobat PDF files. If this file was on your computer, double-clicking the icon in Windows would open that file in Adobe Acrobat or the PDF reader installed on the computer.

Q5: What is HTML ? Its history? What point should be in mind while coding HTML?

HTML is a computer language devised to allow website creation. These websites can then be viewed by anyone else connected to the Internet. It is relatively easy to learn, with the basics being accessible to most people in one sitting; and quite powerful in what it allows you to create. It is constantly undergoing revision and evolution to meet the demands and requirements of the growing Internet

audience under the direction of the » W3C, the organisation charged with designing and maintaining the language.

6. HTML is the standard markup language for creating Web pages.

- HTML stands for Hyper Text Markup Language
- HTML describes the structure of Web pages using markup
- HTML elements are the building blocks of HTML pages
- HTML elements are represented by tags
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so on
- Browsers do not display the HTML tags, but use them to render the content of the page

History

The first version of HTML was written by Tim Berners-Lee in 1993. Since then, there have been many different versions of HTML. The most widely used version throughout the 2000's was HTML 4.01, which became an official standard in December 1999. Most pages on the Web today were built using either HTML 4.01 or XHTML 1.0. However, in recent years, the W3C (in collaboration with another organization, the WHATWG), has been working on a brand new version of HTML, HTML5. Currently (2011), HTML5 is still a draft specification, and is not yet an official standard.

Important Points

HTML consists of a series of short codes typed into a text-file by the site author — these are the tags. The text is then saved as a html file, and viewed through a browser, like *Internet Explorer* or *Netscape Navigator*. This browser reads the file and translates the text into a visible form, hopefully rendering the page as the author had intended. Writing your own HTML entails using tags correctly to create your vision. You can use anything from a rudimentary text-editor to a powerful graphical editor to create HTML pages.

The most important HTML5 tags would be structure tags. Learn how to properly outline your page with the right semantic tags, such as heading, nav, article, section, footer, h1, ect. Then make sure you understand where and why to use these tags. These structure tags will get you through most HTML5 pages.

Q7: Merits and demerits of HTML?

Advantages of HTML:

1. First advantage it is widely used.
2. Every browser supports HTML language.
3. Easy to learn and use.
4. It is by default in every windows so you don't need to purchase extra software.

Disadvantages of HTML:

1. It can create only static and plain pages so if we need dynamic pages then HTML is not useful.
2. Need to write lot of code for making simple webpage.
3. Security features are not good in HTML.
4. If we need to write long code for making a webpage then it produces some complexity.

Disadvantages & Advantages of HTML

Disadvantages	Advantages
•It takes a lot of time to create anything that even resembles a webpage. •It isn't as flexible as other webpage developers like Dreamweaver. • It is not centralised, all the webpages must be edited separately.	•Easy to create a webpage. •It's plain text so it's easy to edit. •Fast to download because the text is compressible. •Can be used to present just about anything on a webpage.

Q: what is map? Types of maps.

A map is an ADT (Abstract Data Type) where key-value pair (k-v) are stored in an array. The 'key' is an identifier for some kind of data, and the 'value' is the content that is being identified or saved. Map data type is ideal to use in look up type situations where there is an identifying value and an actual value that is represented by the identifying value. A few examples where the map data type can be used are:

- Student ID numbers and last names
- House numbers on a street and the number of pets in each house
- Postal codes and names of cities
- Driving licenses and last names

Types of maps

1. **Array:** An array refers to a set of similar elements. It consists of an index. This acts as a pointer to each element of this structure. Unlike most other data storage types, arrays have a fixed length. Its elements are placed in a contiguous fashion. Each of the data item has a fixed memory address.
2. **Stack:** Stack refers to an orderly arrangement of data. It consists of just one end. This end is used for both, data addition as well as removal. It is said to follow the LIFO pattern. 'Push' and 'Pop' are the two functions of this data structure.
3. **Queue:** A queue is a list with a linear pattern which has two ends: front and rear. The front end allows deletion of data items from the list. The rear end allows insertion of data items into the list.
4. **Linked List:** A linked list is a collection of objects which are linked to each other in a linear pattern. Each of these objects is called a node. The first object is called the front node or the head. Each of the nodes store some data in them.
5. **Set:** Finite set is a mathematical concept, based on which the 'set' storage type is implemented. This data structure consists of mathematical functions defining the conditions and elements belonging to it.
6. **Graph:** A graph consists of a bounded set of nodes (also termed as vertices). A pair of two nodes is called an edge. Each node has some data associated with it. Whereas, the edges indicate the pointers between the nodes.
7. **Tree:** A tree consists of a set of nodes which are linked via pointers. Each structure has a root node and branches containing the remaining data nodes.
8. **Hash Table:** Hash table is a type of table whose main function, in addition to data storage, is mapping the keys to values. The table has some keys to be mapped, a hash function, and buckets. Each bucket is an array which stores data.

9. **File:** A file is a collection of multiple records. Each record consists of one or more elements. These elements are called fields. In case of this data structure, every record is assigned a field and a key.

Q: Define mouse

A computer mouse is an input device that is most often used with a personal computer. Moving a mouse along a flat surface can move the on-screen cursor to different items on the screen. Items can be moved or selected by pressing the mouse buttons (called clicking).[1] Today's mice have two buttons, the left button and right button, with a scroll wheel in between the two.

Q: Define narrator

Microsoft Narrator, also called Narrator, is a free screen reader utility included with Microsoft Windows. While not as robust as a full-function screen reader, it can read dialog boxes and window controls in most of the basic applications for Windows. Good examples of Narrator's uses are basic navigation, aiding in the installation of a different screen reader, or as a backup reader during updates.

Q: Define web 2.0

Web 2.0 is the term given to describe a second generation of the World Wide Web that is focused on the ability for people to collaborate and share information online. Web 2.0 basically refers to the transition from static HTML Web pages to a more dynamic Web that is more organized and is based on serving Web applications to users.

Q: Define homepage

A home page is a webpage that serves as the starting point of website. It is the default webpage that loads when you visit a web address that only contains a domain name. For example, visiting <https://techterms.com> will display the Tech Terms home page. The home page is located in the root directory of a website.

Q: Define aliasing

Aliasing describes a situation in which a data location in memory can be accessed through different symbolic names in the program. Thus, modifying the data through one name implicitly modifies the values associated with all aliased names, which may not be expected by the programmer. As a result, aliasing makes it particularly difficult to understand, analyze and optimize programs. Aliasing analyzers intend to make and compute useful information for understanding aliasing in programs.

Q: Define charts and their uses in classroom

Definition: A chart is a group of related facts presented in the form of a diagram, table, graph, or other visually organized model.

Characteristics of Charts:

1. be clear, easy to understand and easy to find.
2. display content that is current and supports complex skills.
3. have a clear purpose.
4. include steps for how to do specific strategies or procedures.
5. have visuals including symbols, pictures, or photos to go with words.

Benefits

Charts are an excellent tool that can help students become independent thinkers and problem solvers when working to master the CCSS. And although the standards are complex, when the processes are broken down and displayed for students, they can more easily internalize and master them. Charts don't have to be perfect! They are most successful when they are created by or with students, and modified or altered as students' understanding of the concept changes.

Q: Benefits of ICT

ICT or Information and Communication Technology is the infrastructure that helps you to enter the digital world! There is no universal definition for ICT; this term is usually used to refer to all devices connected with information and communication technologies. These networking components unite people all over the world, including governments, companies and nonprofit agencies in one network. ICT combines the internet-based and mobile technology spheres. It also includes non-modern technologies like landed telephones, television and radio broadcast.

1. **Resources** This is one of the main benefits of ICT to students and teachers. When they use the internet and communication devices, they can enter a world of endless possibilities and information. Students can learn to become independent and effective learners. Teachers can also use the resources of the internet to make their lectures and classes more interesting.
2. **Love of ICT** New technologies can provide a lot of possibilities for students; however, their main interest in these technologies lies in its entertainment factor. If students can combine their desire for fun with a need for learning, they will become destined for success.
3. **Relevance around global world:** When students use ICT, they connect with all people around the globe. It's necessary for them to stay active in the network to learn about new possibilities. It will be valuable for students to learn about the possibilities connected to their future education or jobs.
4. **Good for the Environment** When you use computers instead of books and notebooks, you save a lot of trees from getting cut down. Deforestation is one of the major problems of humanity. We use too many resources for our needs. That is why ICT provides new possibilities to save trees, and also helps students understand the necessity of nature.
5. **Useful for Disabled Students** Students with health problems can now live in a world of possibilities by using a computer; they can learn and study even at their homes. Therefore, all students can be placed on equal footing to get the necessary education
6. **Self-regulated learning:** Students can become much more independent in learning when they have access to the resources of the global network. A teacher can teach them the basis of how to search for information on the internet, and after that, students may decide which website they should use to get the information they need.
7. **Unity with the world** There is no reason why students should be limited to the borders of their countries. With the help of ICT, they can chat and cooperate with people all over the world. For instance, they can easily learn new languages by communicating with native speakers!
8. **Teacher's community** ICT in the education sector provides benefits for teachers too. They can use forums, communities and other organizations to help their students.

Moreover, it will provide more professionalism in the sphere of pedagogy. It can also develop new ways and methods of studying.

9. **Practical Use:** Students can put their knowledge into practice by using internet and communication technologies. For instance, if they learn how to speak foreign languages, they can try to work as translators. If they study programming, they can try to develop landing pages, apps, etc.
10. **Interactive classes** Teachers may use new technologies to improve their lessons by getting more interesting materials and showing them off! For instance, it is possible to use interactive blackboards to study mathematics, languages, etc.

Q: Why teaching is noble profession?

"A teacher is never an ordinary person. Construction and destruction can be produced in his lap." The above given quotation by Chanakya shows the power of teachers and teaching. Teaching is a very important activity on which the base of any nation and culture is established. So let's see what teaching is.

The word 'teaching' means the work of a teacher to provide knowledge and guidance. It is one of the oldest and noblest services to the society in any culture. But the question arises if teaching is a profession or a duty first. According to The Oxford Dictionary the word 'profession' means 'a paid occupation' and 'duty' means 'a moral responsibility'. So it's better to call teaching a duty than a profession. Let's consider that this noble profession is a duty and discuss its importance then. Teaching

not only shows the right path that the students should follow but also prepares the human resource for the further development of the nation. It has the potential to have a great impact in the molding of the next generation. That is why education should be valued by social institutions like government, the church, the family and civil society. Therefore teaching is:

A process to impart knowledge and information

A process of causing change

A process to instruct and guide others

Teaching is the noblest among all the professions since all professionals underwent education with a teacher. It's also a process to prepare the next generation of skilled professionals and workers like politician, engineers, doctors, policemen, priests, educators, legislators and good citizens. Teaching, therefore, is considered as a means for which God uses a teacher as an instrument to touch lives.

Q: Define virtual learning

Usually the common term used in "Virtual learning environment (VLE)". "A virtual learning environment (VLE) is a set of teaching and learning tools designed to enhance a student's learning experience by including computers and the Internet in the learning process"

Process of learning using any of the VLE is virtual learning.

- Computer-Based learning: Instructions are delivered by the computer or software installed in the computer
- *Internet-Based*: This is similar to computer based learning. But its not the software in the computer but the instructions are delivered through web
- *Remote Teacher Online*: The teachers is not physically present, but delivers the lecture virtually
- *Blended Learning*: This combines traditional face-to-face instruction and use of any other virtual learning environment
- *Facilitated Virtual Learning*: This is *computer-based*, *Internet-based* or *remote teacher online* instruction that is supplemented by a human "facilitator."

Similar forms of virtual learning are sometimes grouped into broader categories:

- *Online Learning*: This is any form of instruction that takes place over the Internet.
- *Full-Time Online*: This is online learning with no regular face-to-face instruction or facilitation.
- eLearning : learning conducted via electronic media, typically on the Internet.

Q: Define WWW world wide web in detail.

In British a vast network of linked hypertext files, stored on computers throughout the world, that can provide a computer user with information on a huge variety of subjects.

In America, a group of Internet sites interconnected by means of hypertext, providing access to images and sound in addition to text.

Q: Define E-Government.

E-Government refers to the use of internet technology as a platform for exchanging information, providing services and transacting with citizens, businesses, and other arms of government. E-Government may be applied by the legislature, judiciary, or administration, in order to improve internal efficiency, the delivery of public services, or processes of democratic governance.

Q: Describe the term database.

A database is a collection of information that is organized so that it can be easily accessed, managed and updated. Data is organized into rows, columns and tables, and it is indexed to make it easier to find relevant information. Data gets updated, expanded and deleted as new information is added. Databases process workloads to create and update themselves, querying the data they contain and running applications against it.

Q: write about software for presentation.

1. Microsoft Office PowerPoint is a presentation program developed for the Microsoft Windows and Mac OS computer operating systems. Businesspeople, educators, and trainers use it often. It is a component of the Microsoft Office system. In PowerPoint, as in most other presentation software, text, graphics, movies, and other objects are positioned on individual pages or "slides". The "slide" analogy is a reference to the slide projector. Slides can be printed, or (more often) displayed on-screen and navigated through at the command of the presenter. Slides can also form the basis of webcasts.
2. Google Slides offers a simplified web-based PowerPoint style tool which can be used to quickly make and share decks. This solution offers a great tool for collaborating with co-founders and advisors, as well as easier sharing via email or directly online.
3. Prezi as the most innovative of the best presentation software companies in 2018.
4. Keynote is designed for creating slides, but it comes with the key advantages of visually stunning templates and a simple, clear interface.

Q: Explain firewall.

A firewall is software used to maintain the security of a private network. Firewalls block unauthorized access to or from private networks and are often employed to prevent unauthorized Web users or illicit software from gaining access to private networks connected to the Internet. A firewall may be implemented using hardware, software, or a combination of both.

Q: Define Wide Area Network and its characteristics.

A wide area network (WAN) is a network that exists over a large-scale geographical area. A WAN connects different smaller networks, including local area networks (LANs) and metro area networks (MANs). This ensures that computers and users in one location can communicate with computers and users in other locations. WAN implementation can be done either with the help of the public transmission system or a private network.

The major features of WANs are listed below :

- Multiple computers are connected together
- It connect devices that are separated by a broader geographical area than a LAN • A WAN usually interconnects multiple LANs
- Communication links between computers are provided by telephone networks, public data networks, satellites etc.
- Links are of low capacity (that is low data rate)
- Bit error rate is higher (1 in 100,000) compared to that for a LAN.
- The data rates of WAN is low as compare to data transfer rate of local area network , and the signal propagation delay is much greater than the local area network. The typical data rates for WAN are 56kbps to 155Mbps, 622Mbps, 2.4 Gbps or higher speed WAN.

Q: Explain the importance of audio visual aids briefly.

Importance of Audio Visual Aids

The importance of audio visual aids is evident when it comes time to teach a new or abstract concept that may be difficult to picture or grasp. Audiovisual aids allow teachers to demonstrate things to students that, otherwise, may not be clearly conveyed. When you lecture to students on a new subject -- like ringtail lemurs' mating habits -- you are limited to verbal descriptions. However, when you can demonstrate something through sight and sound, students pick up more information through the audiovisual experience. For instance, a film clip on ringtail lemurs' mating habits gives students an academic experience they could not obtain from a lecture or a textbook.

1. Reaching Specific Learners

Students have different learning methods. Howard Gardner of Harvard University -- who advanced the multiple intelligences theory, which suggests individuals learn things differently -- identified seven learning styles: visual-spatial, bodily-kinesthetic, musical, interpersonal, intrapersonal, linguistic and

logical. While some students may learn more effectively through hands-on activities, others may learn through musical or visual responses. The importance of AV aids can be seen when classroom audiovisual tools allow musical learners to learn information in a way that stimulates their preferred learning style. Similarly, the importance of charts in the teaching and learning process is clear when introducing concepts to visual learners and students who are more quantitative than verbal.

2. Spicing Up the Curriculum

Experienced teachers have observed first hand the importance of AV aids in keeping students focused on the lesson. Audiovisual aids in the classroom add diversity to an instructor's teaching method. Whether you teach a group of preschoolers or college students, incorporating audiovisual aids into your lessons helps you keep students' attention. On the contrary, too much of one type of teaching method, whether it's lecturing or discussion, can grow dull and cause students to lose interest in the material. Retention of information typically is greater when audiovisual aids complement or present information.

Remembering the importance of charts in the teaching and learning process will create an active learning environment.

3. Promoting Clear Communication

By using audiovisual tools in the classroom, teachers show students how to communicate through various media. Students are introduced to the concept of conveying information creatively -- which comes in handy when they must give presentations in class or later in their careers. Exposure to audiovisual aids in school teaches kids how to communicate through sight and sound, and to tie the things they see and hear together to understand an important message.

Q: How do the exposure and use of ICT in school affects future employment?

Technology is rapidly changing the job market. Will your dream job still be around in ten years' time? What skills are employers currently on the hunt for? Technology. Whether you find that you embrace every new update, or struggle to keep up with the changes, it's effect on our day-to-day lives is unmistakable. Recent technological developments are vastly changing the way we live, and in particular, the way we work. With the job market a competitive and ever-evolving arena, determining your career path is more than just following your passions and skills, it's about being adaptive in this digital age and understanding what skills and jobs are in high demand, and which are facing potential extinction. It's important to try not feel disorientated or uncomfortable with these rapid changes, but instead, change your mindset and embrace the opportunities technology brings, through focusing on the right skills to increase your employability and succeed in the new tech-savvy work scene.

Q: what is browser?

A browser is an application program that provides a way to look at and interact with all the information on the World Wide Web. The word "browser" seems to have originated prior to the Web as a generic term for user interfaces that let you browse (navigate through and read) text files online.

Q: what is Bluetooth?

Bluetooth is an open wireless technology standard for transmitting fixed and mobile electronic device data over short distances. Bluetooth was introduced in 1994 as a wireless substitute for RS-232 cables. Bluetooth communicates with a variety of electronic devices and creates personal networks operating within the unlicensed 2.4 GHz band. Operating range is based on device class. A variety of digital devices use Bluetooth, including MP3 players, mobile and peripheral devices and personal computers.

Q: what is sound card?

A sound card is an expansion component used in computers to receive and send audio. Sound cards are configured and utilized with the help of a software application and a device driver. The input device attached to receive audio data is usually a microphone, while the device used to output audio data is generally speakers or headphones.

Q: What is software?

Software, in its most general sense, is a set of instructions or programs instructing a computer to do specific tasks. Software is a generic term used to describe computer programs. Scripts, applications, programs and a set of instructions are all terms often used to describe software.

Q: What is download accelerator?

A download manager is software tool that manages the downloading of files from the Internet, which may be built into a Web browser, or a usually more sophisticated standalone manager.

Q: What is Trojan Horses?

A Trojan horse is a program that appears harmless, but is, in fact, malicious. Unexpected changes to computer settings and unusual activity, even when the computer should be idle, are strong indications that a Trojan is residing on a computer. A Trojan horse may also be referred to as a Trojan horse virus, but that is technically incorrect.

Q: Define google

Google is an Internet search engine. It uses a proprietary algorithm that's designed to retrieve and order search results to provide the most relevant and dependable sources of data possible. Google's stated mission is to "organize the world's information and make it universally accessible and useful." It is the No.1 search engine in the world, a position that has generated criticism and concern about the power it has to influence the flow of information on the Web.

Q: Define internet?

The internet is a globally connected network system that uses TCP/IP to transmit data via various types of media. The internet is a network of global exchanges – including private, public, business, academic and government networks – connected by guided, wireless and fiber-optic technologies. The terms internet and World Wide Web are often used interchangeably, but they are not exactly the same thing; the internet refers to the global communication system, including hardware and infrastructure, while the web is one of the services communicated over the internet.

Q: What is dialogue box?

A dialog box is a user interface element, a type of window that is used to enable communication and interaction between a user and a software program. The dialog box appears when the program either needs to give the user information in an urgent manner that involves interruption, such as when an error occurs, or if the program requires immediate input or decision from the user, such as when the program closes and needs to know if the changes made have to be saved or not.

Q: What is folder?

A folder is the virtual location for applications, documents, data or other sub-folders. Folders help in storing and organizing files and data in the computer. The term is most commonly used with graphical user interface operating systems.

Q: What is Baud?

Baud (Bd) is a data transmission unit that shows how many signaling elements or symbol changes (electronic state change) are sent per second in a line code or a digitally modulated signal. It is not the measure of data transfer speed, but the measure of modulation. This should not be confused with the actual gross data transfer rate, which is expressed in bits per second. Although the two are related, they are not equal.

Q: What is an IP Address?

An Internet Protocol address (IP address) is a logical numeric address that is assigned to every single computer, printer, switch, router or any other device that is part of a TCP/IPbased network.

The IP address is the core component on which the networking architecture is built; no network exists without it. An IP address is a logical address that is used to uniquely identify every node in the network. Because IP addresses are logical, they can change. They are similar to addresses in a town or city

because the IP address gives the network node an address so that it can communicate with other nodes or networks, just like mail is sent to friends and relatives.

The numerals in an IP address are divided into 2 parts:

- The network part specifies which networks this address belongs to and
- The host part further pinpoints the exact location.

Q: What is Decompression?

Decompression is the process of restoring compressed data to its original form. Data decompression is required in almost all cases of compressed data, including lossy and lossless compression. Similar to compression of data, decompression of data is also based on different algorithms. Decompression is considered important, as compressed data needs to be restored back to standard state for usage. Decompression is widely used in data communications, multimedia, audio, video and file transmissions. Decompression is also known as uncompression.

Q: Describe the three benefits of the E-learning for the teachers and the institutions?

In this context, e-learning is a channel and tool through which teachers can improve their teaching styles; here are some of its key benefits:

- **connecting** – e-learning platforms allow teachers to stay connected to their students outside of school hours in order to exchange resources, videos, ideas, methodologies, and pedagogical practices
- **flexibility** – as e-learning combines a huge variety of different resources such as videos texts, presentations, and quizzes, it gives teachers the opportunity to adapt their teaching methods to the learning styles of their students
- **innovation**– the use of new technologies gives teachers the freedom to experiment in their teaching practice and the ability to make changes if something doesn't work
- **Accessibility** – as online platforms are available 24/7, they allow teachers to develop and create their materials whenever they want to and according to their schedules. It is also important to note that all teaching materials can be reviewed as many times as needed
- **passion-based learning** – online platforms allow teachers to support their students outside the classroom where they don't have much time together, and to give them additional instructions on subjects like arts, singing, chess, and cooking. It gives teachers the opportunity to lead students to success both during and after school hours
- **Efficiency** – e-learning allows teachers to receive constant feedback from their students about whether they understand the material, if it is interesting to them, etc. Online platforms provide a large variety of assessment tools. .
- **Save time** -time is one of the most precious commodities for educators. By eliminating the need to commute to school or university during evenings and weekends, educators can spend more time on learning concepts.
- **greener and cost-effective** – teachers do not have to pay for classroom rentals, travel fees, or printing costs since e-learning activities take place online . Paper consumption is also significantly reduced because all of the information is in an electronic format.
- **Creativity** – on a daily basis, teachers are challenged to become more versatile and creative in order to attract the attention of their students. Using a variety of online resources for an online training provokes and develops the creative thinking processes of educators.
- **Lifelong learning** – e-learning platforms are one of the channels that educators can use to deepen their knowledge and increase their skills.
- **Building a community** – online platforms allow educators to stay connected with colleagues from all over the world, as well as to share ideas and gain inspiration.
- **Self-reflection** – teachers can make audio and video recordings of their lessons and then watch, analyze, and reflect on the recordings afterwards.
- **Observation** – observing other educators in your school or via different online courses can be very enlightening. It helps teachers compare different teaching styles, practices, etc., as well as become more aware of what kind of teachers they are.

- **Feedback** – e-learning platforms are a good opportunity for teachers to receive feedback on their teaching style from their students and colleagues in order to see where they can tweak their habits and make improvements.

Q: Benefits of ELearning for student

1. Online Learning Accommodates Everyone's Needs
2. Lectures Can Be Taken Any Number Of Times
3. Offers Access To Updated Content
4. Quick Delivery Of Lessons
5. ELearning helps in creating and communicating new training, policies, concepts, and ideas. Whether it is for formal education or entertainment, eLearning is very quick way of learning.
6. ELearning enables educators to get a higher degree of coverage to communicate the message in a consistent way for their target audience. This ensures that all learners receive the same type of training with this learning mode.
7. ELearning is cost effective as compared to traditional forms of learning. The reason for this price reduction is because learning through this mode happens quickly and easily. A lot of training time is reduced with respect to trainers, travel, course materials, and accommodation.

Q: Define the meaning of the teaching, and definitions of the teaching and the functions of the teaching?

Meaning of Teaching

The job or profession of a teacher. Something that is taught: the ideas and beliefs that are taught by a person, religion, etc.

Definition of Teaching

According to Gage, "Teaching is a form of interpersonal influence aimed at changing the behavior potential of another person."

Edmund Amidon defined it as-" Teaching is an interactive process, primarily involving class room talk which takes place between teacher and pupil and occurs during certain definable activity."

CONCEPT OF TEACHING:

Teaching is process which usually takes place in the class room situations. It is more of formal processes. In the class room situations we see that the teacher has something in his mind and he wants to convey it to the students. For this purpose, he takes the help of teaching. He makes all efforts to make the students understand it. His teaching is successful if the students are able to grasp it fully.

NEED OF TEACHING:

Through teaching, the teacher aims at

- i. Giving some knowledge to the students;
- ii. Passing some information to them;
- iii. Making the students acquire some skill;
- iv. Changing the attitude of the learners;
- v. Modifying the behavior of the students;
- vi. Giving some experiences of life; etc.,

DEFINITIONS OF TEACHING:

- i. *According to Rabindra Nath Tagore*, "A Teacher can never truly teach unless he is still learning himself. A lamp can never light another lamp unless it continues to burn its own flame."
- ii. *H.C. Morrison (1934)*: "teaching is an intimate contact between a more mature personality and less mature one which is designed to further the education of the latter."
- iii. *N. L. Gage(1962)*: "Teaching is a form of interpersonal influence aimed at changing the behavior potential of another person"
- iv. *B.D. Smith (1961)*: "Teaching is a system of actions intended to induce learning."
- v. *John Dewey*: "One might as well say he has sold when no one has bought, as to say he has taught when no one has learned."

Functions of Teaching:

In teaching process, the active or functional part is played by the independent and dependent variable. Mainly following three types of functions are performed by these variables:

- I. *Diagnostic functions*
- II. *Prescriptive functions*
- III. *Evaluative functions*

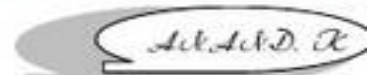
I. Diagnostic functions:

The goal is to bring desirable changes in the behavior of students. The initial task needs a proper diagnosis for the prescription of appropriate treatment (the actual attempts) for bringing desirable behavioral changes in the students. Accordingly, a teacher has to perform the following diagnostic functions:

1. He has to diagnose the entering behavior of the student. The initial potential of the student in terms of cognitive, cognitive and affective abilities should be properly diagnosed with the help of some diagnostic tests.
2. He has to diagnose and formulate specific educational objectives, the type and amount of behavioral changes he wants to introduce in the student in the light of the entering behavior and environmental conditions.
3. He has to analyze the content, instructional material and environmental facilities available for carrying out his task.
4. He has to diagnose his own potential and capabilities and bring desirable improvement in his own behavior for the success of his own mission.

II. Prescriptive functions:

On the basis of diagnosis, the teacher takes decision about the needed prescription for achieving the stipulated objectives. Accomplishment of objectives needs an appropriate interaction between the teacher and the student which, in turn, needs proper management of



the intervening variables by the teacher. Accordingly, he has to perform the following functions;

1. Selecting appropriate contents and organizing them into proper sequence.
2. Selecting proper teaching techniques, strategies and feedback devices in view of the individual difference among the students.
3. Seeking essential cooperation from the students for a purposeful interaction.

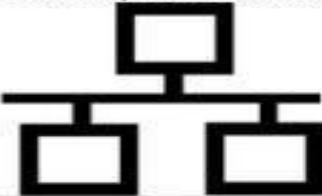
In the performance of prescriptive functions also, the teacher is likely to be more active than the students. The prescription is made for the student to bring desirable changes in his behavior. He has to work for the purposeful interaction and give his sincere cooperation for the teacher in exercising the various prescriptive functions.

III. Evaluative functions:

Evaluative function concerned with the tasks of evaluating the progress and outcomes of the prescriptive functions that may be decided in the form of realization of the stipulated objectives. The failure in the realization of the objectives is essentially a failure in the prescriptive functions either due to improper diagnosis or some serious mistakes in prescribing or carrying out the treatment (actual teaching) task.

Various evaluation devices in the form of tests, observations, interviews, rating scales, inventories and unstructured projective techniques are help in exercising evaluative functions. In contrast to diagnostic or prescriptive functions, the student remains more active in the evaluative functions. He has to respond and evaluate his own progress in terms of the abilities acquired and behavior changes occurred. He is taught and helped by the teacher in bringing improvement in his behavior on the basis of his entry behavior and potentialities. Now, it is this turn to see how far the treatment prescribe for done by the teacher is helpful for him. If the prescription suits him he can go ahead with it. If it does not, he must give his full cooperation to the teacher (just like a patient who has to consult his doctor and seek his advice for further diagnosis and subsequent prescription in order to get maximum advantage for bringing improvement in his behavior.)

Q: how many ways of networking?

Connection type	How it works
Wi-Fi (wireless connection) 	Wi-Fi stands for Wireless Local Area Network. It allows you to connect several computers at once without having Ethernet cables. Wi-Fi is a little broadband zone where radio waves are used to connect devices wirelessly; it is also referred to WLAN. The process for this works in reverse, with the computer sending information to a wireless transmitter.
Ethernet (wired connection) 	Ethernet cable is where the connection transfers to from one signal to another. It is mainly used for local area networking, to get the best connection possible. If an Ethernet cable is damaged it may affect the connection, by disconnecting or running slow.
3G (wireless mobile connection) 	3G works with radio technology within mobile phones. It uses wireless signals from transmitters in the local area. 3G can also be used to connect devices through Wi-Fi, this is called a Wi-Fi hotspot.
4G (wireless mobile connection) 	4G works the exact same way as 3G, but is much faster and is the newer generation connection. Every that is done with 3G is the same as 4G, just slower.
Bluetooth (wireless transfer connection) 	It is a wireless connection that enables you to send and receive files from other devices which are paired together. Bluetooth doesn't require an internet connection and can be used over short distances up to 300 feet. It uses a special radio frequencies that run 2.4GHz band to transmit data. It is a secure way to transfer data and can connect up to eight devices. It is secure because you will need a passcode for pairing together. Bluetooth is a very fast and reliable

Q: coprocessor?

A coprocessor is a supplementary processor unit or an entirely different circuitry that is designed to complement the central processing unit (CPU) of a computer. Its basic functionality is to offload other processor-intensive tasks from the CPU in order to achieve accelerated system performance, by allowing the CPU to focus on tasks essential to the system. There are various types of coprocessors available to perform unique tasks – from I/O interfacing or encryption, string processing, floating-point arithmetic and signal processing.

Q: what is IP?

Internet Protocol (IP) is the principal set (or communications protocol) of digital message formats and rules for exchanging messages between computers across a single network or a series of interconnected networks, using the Internet Protocol Suite (often referred to as TCP/IP). Messages are exchanged as datagrams, also known as data packets or just packets.

Q: What is virus?

Viruses are malicious programs that spread throughout computer files without user knowledge. Most widespread virus infections spread through email message attachments that activate when opened. The vicious cycle of a virus perpetuates as infected emails are forwarded to multiple users. Viruses also spread through shared media, such as Universal Serial Bus (USB) drives.

Q: How many type of graph and its uses?

A Graph is a data structure consisting of vertices and edges. The theory associated with graph is called graph theory.

Types of Graph

1. **Connected Graph:** A graph is said to be connected if there is a path between connecting every pair of nodes or vertices. A graph which is not connected can be divided into connected components call disjoint connected subgraphs.
2. **Disconnected graph:** Graphs can be connected and disconnected. This means that the graph is constructed out of two or more sub-graphs without a path between these components. You can think of a disconnected graph as for the roads of the UK and France, since they aren't connected by land.
3. **TREE Graph:** A graph is a called a tree if it is connected and contain no loop. A tree may contain a special node called the root node of the tree. The exterior nodes of the tree are called leaves of the tree.
4. **Weighted Graph:** A graph is said to be weighted graph if some real number is assigned on every edge of a graph. The weight may be some value, any variable and so on.
5. **Planar and Non-Planar Graphs:** Graph A is planar since no link is overlapping with another. Graph B is non-planar since many links are overlapping. Also, the links of graph B cannot be reconfigured in a manner
6. **Regular graph:** In graph theory, a regular graph is a graph where each vertex has the same number of neighbours; i.e. every vertex has the same degree or valency. A regular directed graph must also satisfy the stronger condition that the indegree and outdegree of each vertex are equal to each other.

Uses of Graphs

- In Computer science graphs are used to represent the flow of computation.
- Google maps uses graphs for building transportation systems, where intersection of two(or more) roads are considered to be a vertex and the road connecting two vertices is considered to be an edge, thus their navigation system is based on the algorithm to calculate the shortest path between two vertices.
- In Facebook, users are considered to be the vertices and if they are friends then there is an edge running between them. Facebook's Friend suggestion algorithm uses graph theory.
- In Operating System, we come across the Resource Allocation Graph where each process and resources are considered to be vertices. Edges are drawn from resources to the allocated process, or from requesting process to the requested resource. If this leads to any formation of a cycle then a deadlock will occur.

Define FTP?

The File Transfer Protocol is a standard network protocol used for the transfer of computer files from a server to a client on a computer network. FTP is built on a client-server model architecture using separate control and data connections between the client and the server

What is Compatibility?

Compatibility is referred to pieces of hardware and/or software which is capable of being used together is described as compatible.

What is BSD?

This is abbreviation of Berkeley Software Distribution. Implementation of the UNIX operating system and its utilities developed and distributed by the University of California at

Berkeley. **BSD** is usually preceded by the version number of the distribution, e.g., **4.3 BSD** is version 4.3 of the Berkeley UNIX distribution. Many Internet hosts run BSD software, and it is the ancestor of many commercial UNIX implementations.

What is BGI?

This is an abbreviation of Binary Gateway Interface. It provides a method of running a program from a Web server. Similar to a Common Gateway Interface (CGI). The BGI uses a binary DLL which is loaded into memory when the server starts. While more efficient than a CGI, the BGI must be compiled and is not easily portable to other environments.

What is workstation?

Client is referred to as a computer that receives services from another computer. A standalone computer on your desk which you use to browse the Web is a client, and the computers from which World Wide Web files are downloaded to your computer are servers. Similarly, a computer (also known as a workstation) connected to a Local Area Network (LAN) is a client that can receive information from and send information to the server that controls the LAN.

Define DCS?

This is abbreviation of Desktop Color Separation. It is an image format consisting of four separate CMYK PostScript files at full resolution, together with a fifth EPS master for placement in documents.

Define Hotlist?

This is a file containing lists of Internet Web pages, enabling you to easily access your favorite Web pages.

Define High key?

It is a light image that is intentionally lacking in shadow detail.

How ICT can be used for illiterate youth?

Issues of equity are central to the discussion of the education-related Millennium Development Goals. For a variety of reasons and in a variety of ways, the introduction of ICTs to benefit education can have profound impact on issues related to Education For All. While for many ICTs offer a promise at greater inclusion of previously marginalized groups (whether marginalized by gender, disability, distance, language, culture, race, age or economic status), their use also brings with it very real dangers of increasing the marginalization of such groups inside the education system.

How should the ICT components of donors be identified and quantified?

If the presence, magnitude and nature of ICT investments in education in developing countries is of interest to donors, standard definitions of ICTs in education must be developed, agreed to, disseminated and utilized, and changes must be made to current coding practices of donor-supported education projects. To this end, the formation of a multi-donor task force to study and this issue and agree upon certain common definitions and coding standards could be useful. Such standards and definitions need not (initially) be comprehensive, but should not focus exclusively on hardware indicators. A comprehensive study of a small number of donor-supported education projects to identify all uses of ICTs components could be a useful tool to support the effort to articulate and agree upon common definitions and standards.

Define BIOS?

BIOS (Basic Input/output System), can be defined as a program which is located in a separate ROM on the motherboard, and defines, as the name suggests, basic input/output system, is a good example.

What is attachment?

A term used in connection with Email. An attachment can be a File of almost any kind - a document file, an image file, a sound file or a video clip - that you can add, i.e. attach, to an email.

define Inline image?

These are graphics that are contained within a document's textual information. In a Web document, these graphics can either be loaded automatically when the page is accessed or loaded manually by clicking on the image's icon.

Define Drum scanner?

Early drum scanners separated scans into CMYK data, recording these directly onto film held on a second rotating drum.

What is Dotted decimal notation? Marks

It is the convention for writing 32-bit IP Addresses as a set of four 8-bit numbers written in base 10 with periods separating them.

What is external viewer?

Program used for presenting graphics, audio and movies while browsing World Wide Web pages via a Web client program. Helper applications are another term for these external programs.

Difference b/w virtual reality and virtual world?3 marks

Virtual Reality

The simulation of an environment by presentation of 3D moving images and associated sounds, giving the user the impression of being able to move around with the simulated environment. Users wear helmets and visors that convey the images and sound and gloves that give them the experience of touching objects. The film Lawnmower Man (1992) focused on a character experiencing virtual reality, albeit with negative consequences.

Virtual World

A type of online three-dimensional imaginary world or game in which participants and players adopt amazing characters or avatars and explore the world, engaging in chat or playing complex games. See Avatar, MMORPG, MUVE. See also Section 14.2, Module 1.5 under the heading Chat rooms, MUDs, MOOs and MUVES. Not to be confused with Virtual Reality, this is a completely different concept.

Differential impact of ICT on subgroups of boys and girls?3 marks

It is well established that ICTs used in education can have differential impact on boys and girls. However, such research has neglected to explore the possible impact of ICTs on certain subgroups of girls (and boys), and the resulting implications for impact of ICT in education on different types of girls and boys.

How private-public sector partnership in introducing ICT in education can cut down the cost of learning??5 marks

Significant work needs to be done related to the costs of ICT in education. All of the claims listed in the related "Knowledge Map" pertaining to costs found in current literature deserve additional scrutiny.

There is little credible data related to the costs of using ICTs to support education in developing countries. Few good, reliable cost studies of ICT in education implementations exist. Those that do exist measure different things, and thus are not easily comparable. While the impact of ICT interventions on student achievement may be difficult to measure, costs explicit and implicit costs of ICT-related educational activities should be measurable.

in 1969 how many universities in Pakistan?

In 1969, we started university, University of Engineering and Technology (UET). At the time there were only two engineering universities in the country; one in the West wing and one in the East

Define GCG? 2

It is a technique for replacing all the neutral tones of an image with an appropriate amount of black.

2. Define Icon? 2

This is a graphic symbol, usually representing a file, folder, disk or tool.

3. what is the meaning of backbone in computer? 2

It is a network used to interconnect several networks together.

4. define Dichroic mirror? 2

Dichroic Mirror

This is a special type of interference filter, which reflects a specific part of the spectrum, whilst transmitting the rest. It is used in scanners to split a beam of light into RGB components.

5. what is computer search service? 2

It is Computer Search Service (CSS), a fee-based service, offers access to more than 500 databases, primarily in the sciences and social sciences, from which the Library can create customized bibliographies.

6. define densitometer? 3

It is a measuring instrument that registers the density of transparent or reflective materials. Colors are read as tonal information.

7. define resampling? 3

It is an increase or reduction in the number of pixels in an image, required to change its resolution without altering its size. See also down-sampling and interpolation.

8. what is FDDI? 3

It is abbreviation of Fiber Distributed Data Interface. It is an emerging standard for network technology based on fiber optics that has been established by the American National Standards Institute (ANSI). FDDI specifies a 100-million bit per second data rate.

9. define header? 3

It is the portion of a packet, preceding the actual data, containing source and destination addresses, error checking and other fields. A header is also the part of an electronic mail message that precedes the body of a message and contains, among other things, the message originator, date and time.

10. what is the meaning of computer program? 3

Computer Program is a set of instructions that the computer carries out in sequence to perform a given task. Programs are written in English-like programming languages (e.g. C, Pascal), and are then converted into binary machine instructions via a compiler or an interpreter.

What is virus?

Virus is a malicious program with the possibility of self-replication. In a computer it seeks other files, trying to infect them and the ultimate goal is to spread to other computers. Depending on what malicious code it contains, it may cause more or less damage to your computer.

What is HTML?

It is called Hyper Text Markup Language. A system for tagging various parts of a Web document that tells the Web client programs how to display the document's text, links, graphics and attached media.

How many type of graph and its uses?

Graphs normally used to represent the nature of the relation of two dependent variables. Different types of graphs are

1. Line graph
2. Bar graph
3. Pie graph

What is Floating point?

Floating-point Processor

It is a special chip that handles sophisticated calculations, such as those used in spreadsheets, CAD, and scientific programs.

Q2: what is halftoning?

Halftoning Factor

It is a multiplication factor (between 1 and 2) applied to output screen ruling to calculate scanning resolution for optimum output quality. This is also known as the halftoning factor.

Q3: coprocessor?

It is a chip designed specifically to handle a particular task, such as math calculations or displaying graphics on-screen. A coprocessor is faster at its specialized function than the main processor is, and it relieves the processor of some work. A coprocessor can reside on the motherboard or be part of an expansion card, as with an accelerator.

Q4: Defragmentation?

Disk Defragmenter

This arranges the blocks of information for a file into adjacent blocks on your hard drive, which may significantly improve the file access times.

Q5: what is IP?

IP

Internet Protocol is the standard that allows dissimilar hosts to connect to each other through the

Internet. This protocol defines the IP datagram as the basic unit of information sent over the Internet. The IP datagram consists of an IP header followed by a message.

Q6: benefits of ICTs in school other than computer class?

Cost-benefit analyses of situating computers outside of classrooms, but rather in school libraries, teacher training institutions and community tele-centres (including those that are school-based), would be quite useful.

Q9: FTP?

It is abbreviation of File Transfer Protocol. The Internet standard high-level protocol for transferring files from one computer to another across the network.

Q10: ARPANET

This is the abbreviation of Advanced Research Projects Agency Network. It is regarded as a pioneering long-haul network funded by ARPA, an agency of the U.S. Department of Defense responsible for the development of new technology for the military. It served as the basis for early networking research as well as a central backbone during the development of the Internet. The ARPANET consisted of individual packet switching computers interconnected by leased lines.

Muhammad Tayyab ilyas

B.ed preparation group

Edu 430 past final papers

1. Effect of emotional, psychological and cultural on use of ICTs?

The impact of ICT use on learners may be most pronounced not on student achievement, but rather on a learner's sense of self and cultural identity.

2. How ICTs help illiterate youth?

Ans: Issues of equity are central to the discussion of the education-related Millennium Development Goals. For a variety of reasons and in a variety of ways, the introduction of ICTs to benefit education can have profound impact on issues related to Education for All. While for many ICTs offer a promise at greater inclusion of previously marginalized groups (whether marginalized by gender, disability, distance, language, culture, race, age or economic status), their use also brings with it very real dangers of increasing the marginalization of such groups inside the education system.

3. Define information technology?

Ans: IT (Information Technology) encompasses all of the technology that we use to collect, process, protect and store information. It refers to hardware, software (computer programs), and computer networks

4. Download?

Ans: Download means to transfer a file from another computer to your own computer

5. ISO?

Ans: It is abbreviation of International Organization for Standardization, the group that developed the OSI protocols.

6. Internet?

Ans: Internet is a huge collection of computer networks that can be communicated with each other. This word is coined from the words 'interconnection' and 'network'. Through this, each computer user can communicate and share common resources with other users.

7. Pixels?

Ans: Pixel is a contraction of picture element.(DESCRIPTION: What you see on a computer Display Screen is made up of thousands of colour pixels or small dots, which can be set according to the user's choice to produce either low resolution output)

8. Host?

Ans: It is the main computer system to which users are connected.

9. Firewall?

Ans: A firewall is a software package that sits between your computer and your Internet connection, keeping an eye on the traffic going to and from

10. Uses of charts in classroom?

Ans: A chart may be defined as a combination of graphics and pictorial media designed for the orderly and logical visualizing of relationship between key fact and ideas

11. Forgot rest?

12. Long ques were from past paper the file is uploaded yest in group files..

13. Define Folder?

Ans: It is an object that can hold other objects, such as other folders and files

14. Define ISO?

Ans: It is abbreviation of International Organization for Standardization, the group that developed the OSI protocols.

15. Define ERIC?

It is an index to educational journal articles and documents collected by national clearing houses for educational information. It provides summaries of the articles and documents, and contains citations back to 1966.

16. Define ICT?

ICT is concerned with the storage, retrieval, manipulation, transmission or reception of digital data

17. Define associate?

It is linking a document with the program that created it so that both can be opened with a single command.

18. Define internet?

Internet is a huge collection of computer networks that can be communicated with each other. This word is coined from the words 'interconnection' and 'network'. Through this, each computer user can communicate and share common resources with other users.

19. Describe the benefits of ICT for the teachers and the institutions?

Educational informatics represents the intersection of three disciplines: teaching and learning, information science, and information communication technologies (ICTs)

Quick access to information • Easy availability of updated data • Connecting Geographically dispersed regions • Catering to the Individual differences • Wider range of communication media • Wider learning opportunities for pupils

20. Benefits of ICTs in school and out of school?

Cost-benefit analyses of situating computers outside of classrooms, but rather in school libraries, teacher training institutions and community tele-centres (including those that are school-based), would be quite useful. Improves engagement. When technology is integrated into lessons, students are expected to be more interested in the subjects they are studying. Improves **knowledge** retention. Encourages individual **learning**. Encourages collaboration. ...

21. Define text Diversity?

Lexical diversity is a measure of how many different words that are used in a text, while lexical density provides a measure of the proportion of lexical items (i.e. nouns, verbs, adjectives and some adverbs) in the text.

22. Finger in computer

Finger is a program that tells you the name associated with an e-mail address. ... To finger another Internet user, you need to have the finger program on your computer or you can go to a finger gateway on the Web and enter the e-mail address. The server at the other end must be set up to handle finger requests.

23. Information technology

the study or use of systems (especially computers and telecommunications) for storing, retrieving, and sending information.

24. IP datagram:

Datagram is a combination of the words data and telegram. Therefore, it is a message containing data that is sent from location to another. ... Datagrams are also called "IP datagrams" since they are used by the Internet protocol (IP). This protocol defines how information is sent between systems over the Internet.

25. Definition hypertext:

a software system allowing extensive cross-referencing between related sections of text and associated graphic material.

26. Definition of firewall

a wall or partition designed to inhibit or prevent the spread of fire.

27. Folder

a folding cover or holder, typically made of stiff paper or card, for storing loose paper.

28. Binary number system . 5. Author discover advantages . If two types of students describe shortly?

In mathematics and digital electronics, a binary number is a number expressed in the base-2 numeral system or binary numeral system, which uses only two

symbols: typically "0" (zero) and "1" (one).

Advantages

David carlos Franklin

Binary is extremely simple to implement. Any system that has an "on" and "off" or "high" and "low" state can be used to encode and/or manipulate data.

Binary is the lowest "base" possible (base 2) and hence any higher counting system can be easily encoded (e.g. decimal, octal, hexadecimal, etc.)

29. Uses of lms

The Top Learning Management Systems Benefits

Organizes eLearning content in one location. ...

Provides unlimited access to eLearning materials. ...

Easily tracks learner progress and performance. ...

Reduces Learning and Development costs. ...

Reduces Learning and Development time.

30. Ensure inclusion

There Is No Diversity Without Inclusion

*Imagine hosting a dinner party.

Every guest has their own set of preferences. Some are vegetarian, while others are vegans. Some have a gluten allergy; others don't like eggs.

The problem is, you have a set menu. Even unknowingly, you have managed to alienate half of your guests.

31. Classroom

A classroom is a learning space in which both children and adults learn. Classrooms are found in educational institutions of all kinds, ranging from preschools to universities, and may also be found in other places where education or training is provided, such as corporations and religious and humanitarian organizations. The classroom provides a space where learning can take place uninterrupted by outside distractions.

32. Professional development for educators:

Teacher professional development is any type of continuing education effort for educators. It's one way teachers can improve their skills and, in turn, boost student outcomes.

Learning can take place in formal or informal settings. Formal settings include conferences, courses, seminars, retreats and workshops. Informal opportunities for teacher professional development include independent research or investigation, peer learning initiatives or even just chatting with a colleague in the staff room.

Professional development for teachers takes place on a number of different levels: district-wide, among teachers in a given school, or even on a classroom or individual basis.

33. What are the benefits of situating ICTs for use in schools outside of computer classroom?

Information and communication technology (ICT) has the potential to enhance participation in educational activities for students with physical disabilities. Even though

incorporating ICTs into teaching and learning in education has become an important issue, it is unclear what evidence research has provided. Aim: To investigate type of ICT items and how ICT is being used by students with

physical disabilities, and describe benefits of ICT use in school activities. Method: A systematic literature search, covering the period 2000-May 2012, was performed in the databases AMED, CINAHL, Eric, OTseeker, Psych Info, PubMed, and Scopus. Data analysis entailed extracting, editing, grouping, and abstracting findings. Results:

A total of 32 articles were included, 16 of which were intervention studies. More than half of the studies concerned students with motor impairments. Type of ICT used differed among impairment groups, and ICT seemed to be especially beneficial for writing, spelling and communication. Conclusion: Even though the review found heterogeneity across the studies students seemed to benefit from ICT use regardless of the type. For future research it is important to highlight intervention studies especially for students with visual, hearing and communication impairments.

34.define header? 3

- It is the portion of a packet, preceding the actual data, containing source and destination addresses, error checking and other fields. A header is also the part of an electronic mail message that precedes the body of a message and contains, among other things, the message originator, date and time.

35.What is the meaning of computer program? 3

- Computer Program is a set of instructions that the computer carries out in sequence to perform a given task. Programs are written in English-like programming languages (e.g. C, Pascal), and are then converted into binary machine instructions via a compiler or an interpreter.

36.How to type of learning strategies be selected by use of ICT upon special need and disadvantaged students? What is the gender in terms difference in this regard? 5

- Research into the types of teaching and learning practices fostered by ICT use (most especially more 'learner centric' pedagogical strategies) and how such pedagogies affect disadvantaged students would also be quite useful.

37.How handheld devices which are enabled 3g to use in support of ICT especially educational professional development and management for teachers? Which

are the best practices are used in this regard? 5

- Handheld devices have one great advantage over many other types of ICTs – they are mobile – and anecdotal evidence suggests that this mobility could provide help meet certain needs of schools and educational systems.

38.How ICT components in education support by donors be quantified and identified? 5

- If the presence, magnitude and nature of ICT investments in education in developing countries is of interest to donors, standard definitions of ICTs in education must be developed, agreed to, disseminated and utilized, and changes must be made to current coding practices of donor supported education projects. To this end, the formation of a multi-donor task force to study and this issue and agree upon certain common definitions and coding standards could be useful. Such standards and definitions need not (initially) be comprehensive, but should not focus exclusively on hardware indicators. A comprehensive study of a small number of donor-supported education projects to identify all uses of ICTs components could be a useful tool to support the effort to articulate and agree upon common definitions and standards.

39.Define hard disk drive.(2)

- Hard Disk Drive (HDD) is a place for permanent data storage (it does not delete/clear when computer shuts down). Its features are: large capacity, faster performance in comparison to optical devices but slower in comparison to RAM and are used for permanent data storage. We can distinguish between internal and external hard drives.

40.Define GB(2)

- It is called Gigabyte. It is a unit of data storage size which represents 10^9 (one billion) characters of information.

41.Define DOS(2)

- This is called as Disk Operating System. The operating system used on IBM personal computers and compatible machines. Dotted Decimal Notation It is the convention for writing 32-bit IP Addresses as a set of four 8-bit numbers written in base 10 with periods separating them.

42.Define cookies(2)

- It is a file sent to a web browser by a web server that is used to record one's activities.

43.Define Crush(3)

44. Define Internet(3)

- Internet is a huge collection of computer networks that can be communicated with each other. This word is coined from the words 'interconnection' and 'network'. Through this, each computer user can communicate and share common resources with other users.

45. Define firewall(3)

- A firewall is a software package that sits between your computer and your Internet connection, keeping an eye on the traffic going to and fro. If anything suspicious appears, such as an unauthorized attempt from a remote computer to write information to your hard disc or to send information from your computer to a remote computer, it will block it and warn you. Firewalls have become essential these days because of the frequent attempts being made by hackers to grab confidential information from computers all around the world, e.g. your bank or credit card details, which may be stored in a file somewhere on your computer.

46. Define IP(2)

- Internet Protocol is the standard that allows dissimilar hosts to connect to each other through the Internet. This protocol defines the IP datagram as the basic unit of information sent over the Internet. The IP datagram consists of an IP header followed by a message.

47. Define IP datagram(3)

- It is the basic unit of information passed across the Internet. An IP Datagram is to the Internet as a hardware packet is to a physical network. It contains a source and destination address along with data. Large messages are broken down into a sequence of IP Datagrams.

48. What is the meaning of bookmark?(5)

- Book marks can be defined as, a collection of direct links to predefined web pages which are stored in your web browser. • Bookmarks can be created and managed by the user themselves. Most browsers come packaged with several existing bookmarks.

49. What are the Advantages of Projected aids?(5)

- Teacher can prepare material in advance. • Saves time in classroom. • Provides greater enjoyment in learning • Stimulates more effective learning • Increases retention • Increases attention • Enlarges or reduces actual size of objects. • Can provide multi-sensory/multimedia approach. • Brings distant past and the present into the classroom. • Can make class/presentation more effective.

50. Ict in education fields(5)

- Quick access to information • Easy availability of updated data • Connecting Geographically dispersed regions • Catering to the Individual differences • Wider range of communication media • Wider learning opportunities for pupils

51. Instant feedback and summative assessment(5)

- Instant feedback is the central function of Formative assessment. It typically involves a focus on the detailed content of what is being learned rather than simply a test score or other measurement of how far a student is falling short of expected standards. Nicole and Rick?, synthesizing from the literature list seven principle of good feedback practice: • It clarifies what good performance is; goals, criteria and expected standards - this function cannot be replicated by technology and still requires human intervention. • It facilitates the development of self-assessment in learning – as I narrated earlier the computer science students at our university has serious deficiencies in the English language, in terms of comprehension, vocabulary, writing and speaking. Teaching MUHAMMAD IMRAN 238 them in medium is terribly difficult. Self-assessment is required, which is done through technology based interactive learning application which asses their level of proficiency and gradually build upon it, through instant feedback. • It provides high quality information to students about their learning – this is done both for the teachers and students. • It encourages teacher and peer dialogue around learning. • It encourages positive motivational beliefs and self-esteem. • It provides opportunities to close the gap between current and desired performance – this cannot be achieved through the traditional lecturing style. • It provides information to the teachers that can help shape teaching and it can also help them assess – the system allows for all to be certified at a certain proficiency level, despite varying times contingent upon effort put in. We had dealt specifically with English education since crucial for collegiate education, whatever the subject. Similarly, in mathematics there has been an enormous amount of research in terms of technology and pedagogy; especially in context of information technology in education. Globally, it is understood that whatever the discipline a certain basic proficiency in mathematics is required; probability, risk analysis, statistics, etc. In the slide you see that mathematical knowledge is essential: “it is important for teachers to see how their students approach the problems.” The criticality of learning from one’s mistakes cannot be emphasized. Also, the onus is upon teachers to discover what approach they are adopting; the traditional bookish approach or one which takes into account real problem solving based upon their individual and innovative approaches. “How much

mathematical knowledge and at what levels students use when solving problems.” Grades should not be the sole criteria rather learning in itself should be an objective. “That is knowing how students think in the process of learning or problem solving and that makes it possible for teachers to keep their students overcome conceptual difficulties and in return improve learning in that sense formative assessment is diagnostic .” One approach is to make students make mistakes, which allows for greater conceptual understanding. “In math’s education to employ formative assessment in the classroom a teacher has to make sure that each student participates in the learning process by expressing their idea – there is a trustful environment.” Again a critical point which is related to our actual case studies and one the grievances students expressed at our college. “A friendly environment in which students can provide each other feedback: the teacher also provides feedback. All the instruction is modified according to student needs.” A fundamental change since the teachers is forced to change his teaching style in conformity to student needs – a dynamic change. The inherent feedback provides the requisite clues in adaptation of teaching styles. “In math’s classes thought revealing activity such as modern eliciting activities and generative activities provide good opportunities for covering these aspects of formative assessment.” Turning to an example of formative assessment and instant feedback, which as you know run in parallel, but for formative assessment a better term might be formative learning, since we need to look at it from a higher perspective. This is real example based upon my experience with the science of learning for the past fifteen years and also leads to the use of technology.

52.Q9. Briefly define why ICT benefit for illiterate youth?

Ans: New positive changes in the progressive country and advantage of ICT in education which students experience now a days. The information and communication that one of the main sources of information and that make the life a more efficient. Here are some of the 10 advantage of ict in education

1. Teaching Makes Easier
2. Lesson can be seen, and hear
3. Easier to understand
4. Improve communication
5. Enjoying while learning

6. Students perform better
7. Know more information
8. Improve visual skills
9. Students motivate to go to school
10. Getting a good grade

53.Q10. How can educational content for dissemination through ICTs

Ans. Organization used ICT to capture, store disseminate and allow for the re-use of knowledge. This approach allows many people to search for and retrieve codified knowledge without having to contact the person who originally developed.

54. Define BIOS?

Ans: BIOS (basic input/output system) is the program a computer's microprocessor uses to start the computer system after it is powered on. It also manages data flow between the computer's operating system (OS) and attached devices, such as the hard disk, video adapter, keyboard, mouse and printer.

55. What is attachment?

Ans: Attachment theory is a psychological, evolutionary and ethological theory concerning relationships between humans.

56. define Inline image?

Ans: An inline image is a graphic image that is inserted into a web page to add visual appeal, provide information or to represent a hyperlink. Inline images are usually in GIF, JPEG or PNG formats. They are incorporated into a web page using the data URI scheme.

57. Define Drum scanner?

Ans: Drum Scanner Definition. Drum scanner is an optical scanner designed for use with a computer. In this device, the original image etc

58. What is Dotted decimal notation?

Ans: Dot-decimal notation is a presentation format for numerical data expressed as a string of decimal numbers each separated by a full stop. For example, the hexadecimal number 0xFF000000 may be expressed in dot-decimal notation as 255.0.

68: What is external viewer?

Ans: A program used for presenting graphics, audio and video files. Programs that allow the viewing of GIF and JPEG files and the hearing of AU files fall into this category.

59. Difference b/w virtual reality and virtual world? 3 marks

Ans: Vulnerability

Forced first person view

The 3D effect is natural

Forced first person view

No distractions

Isolation

Vulnerability

60. Differential impact of ICT on subgroups of boys and girls? 3 marks

Ans:

61. How private-public sector partnership in introducing ICT in education can cut down the cost of learning? 5 marks

Ans:

62. Describe How Handheld devices effective in education (especially for the teacher professional development and administration)

Ans:

63. Define GCG? 2

Ans: GCG. Genetics Computer Group. GCG. Garden City Group (various locations)

64. Define Icon? 2

Ans: An icon is a small graphical representation of a program or file. ... Icons help users quickly identify the type of file represented by the icon. The image is an example of "My Computer" icons in different versions of Microsoft Windows.

65. What is the meaning of backbone in computer? 2

Ans: A backbone is a part of computer network that interconnects various pieces of network, providing a path for the exchange of information between different LANs or subnetworks. A backbone can tie together diverse networks in the same building, in different buildings in a campus environment, or over wide areas.

66.define Dichroic mirror? 2

Ans: A dichroic mirror is a mirror with significantly different reflection or transmission properties at two different wavelengths. This relates to one of two possible meanings of the term dichroism. Dichroic mirrors are applied for different purposes.

67.what is computer search service? 2

Ans: A computer programme that searches document, esp. On the world wide web, for a spiced word or phrase and provides a list of documents in which this word or phrase is found

68.define densitometer? 3

Ans: A densitometer is a device that measures the degree of darkness (the optical density) of a photographic or semi-transparent material or of a reflecting surface. The densitometer is basically a light source aimed at a photoelectric cell.

69.define resampling? 3

Ans: Resampling, in terms of audio files, is also known as Sample Rate Conversion. This is done when you need to convert a digital audio file from a given sample rate into a different sample rate.

70.what is FDDI? 3

Ans: Fibber Distributed Data Interface (FDDI) is a standard for data transmission in a local area network.

71.define header? 3

Ans: A header or heading is text at the top of a page in an electronic document or hard copy. ... For example, in Microsoft Word, a header could be created in a document to display the page number of each page.

72.professional development for educators?

Types of field trips :

Excursions usually involve a tour by a person or a group of person to some selected place. An excursion undertaken for gathering objective based learning experience is study tour. The tour made by this group is known as field trips.

73. Define compression:

Codec is also called a compressor, a compression / decompression software component which translates video between its uncompressed form and the compressed form in which it is stored on media (disk, etc.). The two most commonly used video codecs for CD-ROM video are Cinepak and Indeo.

74. Define information technology:

it includes matters concerned with the furtherance of computer science and technology, design, development, installation and implementation of information systems and applications.

75. what is E-mail:

E-mail is the transmission of text messages via Internet. Using attachment, users can exchange files like images, multimedia, documents, etc. E-mail addresses consist of a username, @ sign ("at") and a domain name, e.g. mark.marich@ITdesk.info, where "mark.marich" is the username and "ITdesk.info" is the domain name

76. What is ISO?

It is abbreviation of International Organization for Standardization, the group that developed the OSI protocols.

77. Define BIOS: This is abbreviation of Basic Input Output System. Part of the computer's operating system that is built into the machine, rather than read from a disk drive at startup. 3 marks Questions

78. Differentiate between teaching, training and instructions.

79. what is bps?

It may be defined as, Bits Per Second is the unit used for measuring line speed, the number of information units transmitted per second.

80. what is IP address?

Network addresses are usually of two types:

(1) the physical or hardware address of a network interface card; for Ethernet this 48-bit address might be 0260.8C00.7666. The hardware address is used to

forward packets within a physical network.

(2) The logical or IP Address is used to facilitate moving data between physical networks and is made up of a network number, a subnetwork number, and a host number. All Internet addresses at SDSU have a network number of 130.191, a subnet number in the range of 1-254, and a host number in the range of 1-254.29- Define hacker?

81. Define filter:

It is a piece of software that an application uses for file-format conversion or special effects. PageMaker, for example, has a filter that lets it import Microsoft Word files, while Photoshop has dozens of filters for special effects (such as image blurring). Filters can be part of the main application or external programs called plug-ins.

Long Question 5 marks

82. what do you know about Technology integration. Discuss in detail.

83. How mathematical mindset can use technology and pedagogy for teaching purpose.

"Every time a student makes a mistake in mathematics: they grow in knowledge; they move forward in learning something important; they move a step higher towards knowledge." While what we do is to fail them. When he made this statement at Stanford, a hush fell over the entire room. Most thought this was entirely absurd. "One reason it is so significant that it speaks to the huge power and value of mistakes, although students everywhere think when they make a mistake they are not a math's person or worse that they are not smart in mathematics. Many good teachers have told students for years, that mistakes are useful and that they show we are learning. New evidence on the brain and mistakes says something much more significant."

84. Use of ICTs for professional development of teachers

The impact of ICT use in school and student exposure to ICTs, and the nature of use and

exposure, on student employability in developing has not been well documented. Tracer studies of the impact of ICTs on further study and employment would be useful, as this could be a useful additional measure of educational quality, beyond standardized testing

results.

effect of emotional, psychological and cultural on use of ICT show ICTs help

illiterate youth. ICTs to benefit education can have profound impact on issues related to Education For All. While for many ICTs offer a promise at greater inclusion of previously marginalized groups (whether marginalized by gender, disability, distance, language, culture, race, age or economic status), their use also brings with it very real dangers of increasing the marginalization of such groups inside the education system.

85. Define information technology:

It includes matters concerned with the furtherance of computer science and technology, design, development, installation and implementation of information systems and applications.

86. Download:

Download means to transfer a file from another computer to your own computer. Downloading is the process of copying files to your computer from a remote location. For example, you might download a file from your Web Hosting account to your home PC.

87. ISO:

It is abbreviation of International Organization for Standardization, the group that developed the OSI protocols.

88. Internet:

Internet (network of all networks) is a global system comprised of interconnected computers and computer networks, which communicate by means of using TCP/IP protocols.

89. Pixels:

Pixel is a contraction of picture element. What you see on a computer Display Screen is made up of thousands of colored pixels or small dots, which can be set according to the user's choice to produce either low resolution output, medium resolution output or high resolution output

90. Host:

A company that provides this service is known as a host; and the service it provides is hosting.

91. Firewall:

A firewall is a software package that sits between your computer and your Internet connection, keeping an eye on the traffic going to and fro.

92. Uses of charts in classroom:

This helps us in diagnosing or mapping the process where an individual might falter. It is no longer sufficient to say that a student is not able to grasp a particular concept

93. What is the meaning of computer program? 3
94. How to type of learning strategies be selected by use of ICT upon special need and disadvantaged students? What is the gender in terms difference in this regard? 5
95. How handheld devices which are enabled 3g to use in support of ICT especially educational professional development and management for teachers? Which are the best practices are used in this regard? 5
96. How ICT components in education support by donors be quantified and identified? 5
97. Define hard disk drive.(2)
98. Define GB(2)
99. Define DOS(2)
100. Define cookies(2)
101. Define Crush(3)
102. Define Internet(3)
103. Define firewall(3)
104. Define IP(2)
105. Define IP datagram(3)
106. What is the meaning of bookmark?(5)
107. What are the Advantages of Projected aids?(5)
108. Ict in edcation fields(5)

109. Instant feedback and summative assessment(5)

Computer Network

Inter connected collection of autonomous computers

Definition of ICT

ICT is concerned with the storage, retrieval, manipulation, transmission or reception of digital data.

Communication Technology

- Communication Technology implies the knowledge, skills and understanding needed to exchange information verbally or nonverbally.
- It is processing of information in terms of accessing information, decoding information and sending it via a medium.
- Communication Technology is the electronic systems used for communication between individuals or groups.

Information and Communication Technology

"ICT is defined, as a diverse set of technological tools and resources used to communicate, and to create, disseminate, store, and manage information" (Mangal, 1997).

Characteristics of ICT

- ICT is the integration of Hardware and Software Technology.
- ICT fosters the dissemination of information and knowledge.
- The nature of ICT is digital and virtual.
- ICT is global, anywhere, any time.
- ICT is a key enabler in the creation of networks.

Computer Informatics

- Informatics' is the study of information processing
- Educational Informatics is the study of the application of digital technologies and techniques to the use and communication of information in learning and education.

Educational Informatics is the development, use, and evaluation of digital systems that use pedagogical knowledge to engage in or facilitate resource discovery in order to support learning.

- Studying the techniques and impact of 'digital technologies' within the context of the 'use and communication of information' in educational settings.
- Educational informatics represents the intersection of three disciplines: teaching and learning, information science, and information communication technologies (ICTs)

Computer Networking

- Copper wire
- Microwaves
- Fiber optics
- Communication satellites

Local Area Network

Local area networks are networks within a single building or campus or up to a few kilometers

Characteristics of LAN(Local area network)

- Low delay
- Very few errors
- Higher speed
- LANs are restricted in size (Functions within 10 KMs).
- Transmission time is bounded.
- Simple network management

Metropolitan Area Network (MAN)

- MAN is an extension of LANs.
- It normally covers the area of a city (are less than 100KMs.
- Uses a combination of different hardware and transmission media to cover distances efficiently.

Wide Area Network and its Characteristics

It is defined as networking in a large geographical area, often a country or continent.

Characteristics of WAN

- WAN uses telephone lines or satellite links as a medium.
- LAN may be connected to WAN.
- It covers different cities or countries.

WANs are not owned by any one organization but are under collective ownership and management.

- Complex network design
- Error rate is large.
- Speed is less compared to LAN

Advantages of ICT in Education

- Quick access to information
- Easy availability of updated data
- Connecting Geographically dispersed regions
- Catering to the Individual differences
- Wider range of communication media
- Wider learning opportunities for pupils

Internet

Internet is a huge collection of computer networks that can be communicated with each other. This word is coined from the words 'interconnection' and 'network'. Through this, each computer user can communicate and share common resources with other users.

Web Server

It is a programme and a computer clubbed into one entity which responds to the requests from web browsers for internet resources. The primary function of a web server is to deliver web pages to clients' e. g. Apache HTTP Server, Microsoft Internet Information Services (IIS), Sun Java System Web Server.

Web Browser

It is a programme that allows a user to display and interact with a 'hypertext' documents e. g. Internet Explorer, Net Scape Navigator, Mozilla Firefox, Google Chrome.

Difference between Website and Web Page

Website

Website is defined as a collection of related pages on particular information.

Web Page

It is a file of information stored on a web server.

Homepage

Home page is the first hypertext document to be shown when a user follows a link to the web server.

Search Engines

Search engines can be defined as special tool to search for information on the net.

- AltaVista
- Yahoo
- Google
- Bing
- AOL

World Wide Web (WWW)

- WWW is a vast collection of documents stored on internet computers.
- It was invented by Tim Berner Lee.
- It provides easy access to information.
- It is a system of media rich documents spread across the internet, linked with each other.
- WWW documents contain texts, pictures, sound, video, etc.

Intranet and its Characteristics

Intranet can be defined as collection of private computer networks within an organization.

- Internal or private network of an organization based on internet technology (such as hypertext and protocols) and accessed over the internet.

An intranet is meant for the exclusive use of the organization and its associates (customers, employees, members, suppliers, etc.) and is protected from unauthorized access with security systems such as firewalls.

- An intranet should be closed off to the general public and require a login for authorized users.

Difference between Uploads and Downloads

Both upload and download refer to the process of transferring a file.

Uploading

- Upload means to transfer a file or files from your own computer to another computer.
- Uploading is the process of copying files from your computer to a remote location. For example, you might transfer a file from your home PC to the Yahoo! computer that stores your Web Hosting files.

Downloading

- Download means to transfer a file from another computer to your own computer.
- Downloading is the process of copying files to your computer from a remote location.

For example, you might download a file from your Web Hosting account to your home PC.

Hosting

- For a website to be viewed by other people it must be stored on a computer (server) that is connected to the internet.
- A company that provides this service is known as a host; and the service it provides is hosting.

Uses of Computer Networks

- For sharing resources
- For high reliability
- For saving money

For using as communication medium

- For accessing remote information
- For person to person communication
- For interactive entertainment

Web Resources

Web resources are defined as web pages and documents on the Internet that provide

useful information.

Bookmarks

- Book marks can be defined as, a collection of direct links to predefined web pages which are stored in your web browser.
- Bookmarks can be created and managed by the user themselves. Most browsers come packaged with several existing bookmarks.

Hypertext

- Hypertext is generally the text displayed on a computer display or other electronic device.
- It may have links to other text which the reader can immediately access.

The hypertext pages are interconnected by hyperlinks, typically activated by a mouse click, key press or by touching the screen.

- Hypertext is sometimes used to describe tables, images and other with hyperlinks.
- Hypertext is the underlying concept defining the structure of the World Wide Web with pages often written in the HTML.
- It enables an easy to use and flexible connection and sharing of information over the Internet.

Markup Language

- Markup language is a modern system for annotating a document in a way that is distinguishable from the text.
- The idea and terminology evolved from the "marking up" of paper manuscripts.
- A widely used markup language is Hyper Text Markup Language (HTML), one of the document formats of the World Wide Web.

Learning Aids

Learning aids can be defined as any device/Aids which help self-learning/group learning.

Teaching Aids

- Teaching aids are the physical tools used to convey information in the classroom.
- Teaching Aids are any device which can be used to make the communication more effective.
- Teaching Aids are any device which can be used to make the learning experience more

concrete, more realistic and more dynamic.

- 50 % of what we HEAR and SEE
- 70 % of what we SAY
- 90 % of what we SAY and DO

Psychology of Teaching Aids

Research done by COBUN (1968) indicates that generally we LEARN:

- 1 % through TASTE
- 1.5 % through TOUCH
- 3.5 % through SMELL
- 11 % through HEARING
- 83 % through SIGHT

Research done by COBUN (1968) indicates that generally we remember:

- 10 % of what we READ
- 20 % of what we HEAR
- 30 % of what we SEE

Advantages of Teaching Aids

- Help to learn more and remember for long.
- Motivate the learner.
- Give clarity to learning.
- Give reality and vividness to learning situations.
- Make abstract concrete
- Reduce verbalism.
- Provide variety in methods of teaching and learning.
- Make learning interesting meaningful and permanent.
- Develop deeper understanding.
- Arouse curiosity and self-activity.
- Saving of energy and time
- Spread of education on a mass scale.

Characteristics of Good Teaching Aids

- They should be meaningful and purposeful
- They should be accurate in every respect
- They should be simple
- They should be cheap
- As far as possible, they should be improvised.
- They should be large enough to be properly seen by the students for whom they are meant.
- They should be up to date
- They should be easily portable
- They should be according to the mental level of the student.

They should be motivating the learner.

Programmed Learning

- Programmed learning is a self-instructional and self-corrective technique in which all the learning material is presented to the learner stage by stage through sequentially arranged smaller units called frames. The frames are graded according to the level of difficulty as well as logical sequence.
- Programmed Learning is a carefully specified, systematically planned, empirically established, skillfully arranged and effectively controlled self-instructional technique for providing individualized instruction or learning experience to the learner.
- Programmed learning is available in the form of books, cards, machine, linear programming, branched programming etc.

Hyper Text Markup Language (HTM)

Hyper Text Markup Language (HTML) is a standard language used for developing web pages’.

Hyper Text Markup Language is a markup language that web browsers use to interpret and compose text, images and other material into visual web pages.

History of HTML

- In 1980, a markup language, i.e., Standard Generalized Markup Language or SGML was developed.
- HTML was evolved from SGML.

- Presently, HTML is advanced to DHTML and then to XML (Extensible Markup Language) and now to VRML (Virtual Reality Modeling Language).

HTML Coding

- HTML codes are known as 'HTML Tags'.

An HTML document has two parts, Markup Language Tags and Web page content Text Information

Points to Remember While Writing HTML Document

- Each tag is enclosed within a '<' (left bracket) and '>' (right bracket). This differentiates them from text.
- An opened tag must be closed
- Opening tag is denoted by < >
- Closing tag is denoted by </ >
- Generally the HTML tags are written in upper case
- It is not a case sensitive language.
- Coding can be done in any text Editor (Windows notepad, Ubuntu –G-Edit).
- When we save HTML file, we should add .html or .HTM after the file name.

Teaching as a Noble Profession

- Teaching is regarded as the noblest of all professions in the world.
- A good teacher not only shows the right path that the students should follow but also prepares the human resource for the further development of the nation.
- Teaching has the power to create a healthy and peaceful world.
- It has the capacity to inspire lifelong learning and a passion for knowledge, understanding, and innovation.

Teaching

- "Teaching is an intimate contact between a more mature personality and less mature one which is designed to further the education of the latter."
- The aim of teaching is to transform a total man and his personality

Training

- In training the various kinds of skill are taught.

- Training has specific goals of improving one's capacity, and performance.
- The trainees are taught to perform skills which they attain during training and practice.
- There is mental and physical activity in training, but more and more of the activity is physical.

Instruction

- Instruction is at the higher level of training.
- Instructions tell people how to perform a process.
- In instruction there is better level of the use of intellectual powers of the trainee.
- The aim of teaching is to transform a total man and his personality, but the aim of instruction is to develop the intellectual powers in a given direction.

Computer Assisted Instruction (CAI)

- A self-learning technique, usually offline/online, involving interaction of the student with programmed instructional materials.
- “It is the use of computer to assist in the presentation of instructional material to a student to monitor learning process or to select additional instructional material in accordance with the needs of individual learners”
- CAI facilitates instant access to accurate information with infinite opportunity for practice. It also provides opportunity for systematically organized learning, to the maximum level possible, for all learners. It makes learning easy and interesting.
- Computer Assisted Instruction (CAI) is an interactive instructional technique whereby a computer is used to present the instructional material.
- CAI uses a combination of text, graphics, sound and video in enhancing the learning process.

Teaching Machines

- These were developed by S.L. Pressy.
- It's a piece of device designed to be operated by an individual student for self-learning.
- The student is presented with a question by some form of display on the machine.
- The student is required to respond either to write answer or put a button to indicate tCorrect response.

- The student will inform the correctness of the answer.
- An account is kept of the responses made by the student.

Information and Communication Technology (ICT)

This concept involves transfer and use of all kinds of information. ICT is the foundation of economy and a driving force of social changes in the 21st century. Distance is no longer an issue when it comes to accessing information; for example, working-from-home, distance learning, e-banking, and e-government are now possible from any place with an Internet connection and a computing device.

Information Technology

IT (Information Technology) encompasses all of the technology that we use to collect, process, protect and store information. It refers to hardware, software (computer programs), and computer networks.

Hardware

The concept of hardware includes computer components, the physical and tangible parts of the computer, i.e., electrical, electronic and mechanical parts which comprise a computer.

Working Principle of a Computer

Computer working principle: data are entered into a computer via input devices, then are processed and stored in a system unit, and are finally displayed by the output device.

Personal Computer

Personal computer (PC), as the name suggests, is intended for personal use, as opposed to the server, which is used by a larger number of people simultaneously, from different locations, often via terminals. If you do not intend to move your computer frequently from one place to another, and at the same time you want maximal price/performance ratio, then you should use a desktop computer. In comparison to laptops or tablet computers, it is much larger in size, inconvenient to carry/move, consumes more electricity but has a much better price/performance ratio. Also, they are much easier to upgrade.

Desktop Computers and Laptop or Tablet PC

- Laptop or tablet PC is used by individuals who have the need to travel with a computer or

simply use them for aesthetic reasons when computing power is not an issue. Laptop computers, as opposed to tablet PCs, more closely resemble a personal computer when it comes to data input. Data entry is done via keyboard and mouse, while the tablet PC data entry is done via touch screen.

- Unlike desktop computers, notebooks and tablet PCs are optimized for portability, low power requirements at the expense of performance and can be used (for a limited period of time-i.e. until the batteries are depleted) without connection to the power grid. In order to prepare a laptop or a tablet computer for use without a power connection, it is necessary to recharge the batteries.

Meaning of Teaching

- Teaching is to cause the child to learn and acquire the desired knowledge, skills and also desired ways of living in the society.
- Teaching is the stimulation, guidance, direction and encouragement of learning.
- Teaching is the communication between two or more persons, who influence each other by their ideas and learn something in the process of interaction.
- Teaching is to fill the minds of the learner by information and knowledge of facts for future use.
- Teaching is the process in which learner, teacher and other variables are organized in a systematic way to attain some predetermined goals.

Definitions of Teaching

- "Teaching is an intimate contact between a more mature personality and less mature one which is designed to further the education of the latter." H.C.Morrison (1934)
- "Teaching means many different things, that teaching act varies from person to person and from situation to situation." (Bar, 1961)
- The behavior or activities of persons as they go about doing whatever is required of teachers, particularly those activities which are concerned with the guidance or direction of learning of others." (Ryan, 1965)
- Teaching is the arrangement of contingencies of reinforcement under which students learn. They learn without teaching in their natural environment, but teachers arrange

special contingencies which expedite learning and hastening the appearance of behavior which would otherwise be acquired slowly or making scene of the appearance of behavior which might otherwise never occur." (B.F. Skinner. 1968)

- "Teaching is an act of interpersonal influence aimed at changing the ways in which other persons can or will behave." (N.L. Gage, 1963)

Functions of Teaching

Preservative Function of Teaching

- The main function of TEACHING is to preserve and protect the old values, customs, beliefs, traditions, etc.
- Teaching must help in preserving the vital elements of our heritage.

Transmitive Function of Teaching

- Teaching transfers knowledge, ideologies, theories, principles and attributes from one generation to another. Thus, it contributes greatly to the general progress of society.

Progressive Function of Teaching

- Teaching provides opportunities to recognize and restructure human experiences and as a result of this, there is development of human civilization and culture.

PDA (Personal Digital Assistant) PALM

PDA Personal Digital Assistant (PALM) is a convenient small sized computer. It easily connects to mobile phones and can prove a good solution for less demanding users. As the name suggests, it is a device that fits in the user's palm. Its name directly tells us that this computer is more of an assistant and not a workstation-whose name suggests the superiority in capabilities and computing power, especially in comparison with PDA.

Mobile Phones and Smart Phones

- Mobile phone is a portable electronic device used for distant communication. In recent years, mobile phone has evolved from simple communication device into a multi-functional device. Additional functions, such as short text messaging (SMS), electronic mail, Internet access, contact registration, calculator, clock, alarm, recording and photograph displaying, recording and playback of video clips, sending/ receiving multimedia messages (MMS), audio recording and playback, etc. has turned the mobile

phone into an extremely useful device, whose absence would make active involvement and participation in a modern society not possible.

- Smartphone is a device that merges functionality of phones, PDAs, cameras, camcorders and computers. To function properly, Smart phones use operating systems, which are the basis for application development. Some smart phones can be connected to an external screen and keypad, which creates a working environment, similar to that of a laptop or a desktop computer. Some operating systems for Smartphone are: Google Android, Symbian, Blackberry, Palm Pilot, and Windows Phone.

Unit

The system unit (case) contains a computer's vital parts. There are two basic types of cases:

- Desktop casing is placed on a desk in a horizontal orientation.
- Towers come in 3 sizes (mini-tower, mid-tower and full-tower) and it is vertically orientated.

Mother Board and Protocol

- Motherboard, MBO is computer's basic circuit, to which all computer components are connected, directly or indirectly. Devices are connected to the motherboard through a system bus. System bus connects all devices, ensures data flow and communication between different devices using predefined protocols.
- Protocol describes a manner in which communication between devices is defined. It enables them to address each other and defines how they should look for each other on either system bus or network. Buses can, according to the purpose, be divided into:

1. Serial USB, Firmware, etc.
2. Parallel AGP, PCI, etc.
3. Mixed-Hyper Transport, Infini Band, PCI, etc.

Cache

Cache is a small capacity memory which allows quick access to data. By storing data from working memory in cache, the speed of communication between processor and RAM is increased. Microprocessors use three levels of fast cache, L1, L2 and L3, used to store often used data.

Central Processing Unit

Central Processing Unit (CPU or processor) is a central part of a computer (and can be referred to as the computer's "brain"). It manages all other computer parts, monitors their mutual communication and performs arithmetic-logical operations. Processor speed is measured in hertz (or megahertz or gigahertz). Most famous manufacturers for personal computer processors are Intel and AMD.

Hard Disk Drive

Hard Disk Drive (HDD) is a place for permanent data storage (it does not delete/clear when computer shuts down). Its features are: large capacity, faster performance in comparison to optical devices but slower in comparison to RAM and are used for permanent data storage. We can distinguish between internal and external hard drives.

Read Only Memory (ROM) and Random Access Memory (RAM)

- ROM (Read Only Memory) is a type of permanent, internal memory that is used solely for reading. All BIOS programs are stored in..

.ROM (Random Access Memory) is a working memory in which analyzed data and programs are stored, while computer runs. It allows reading and writing data, and it is deleted/cleared when the computer shuts down.

BIOS

BIOS (Basic Input/output System), can be defined as a program which is located in a separate ROM on the motherboard, and defines, as the name suggests, basic input/output system, is a good example.

Floppy Disk Drive

Floppy Disk Drive is used for storing and reading data stored on a floppy disk. Disk capacity is 1.44MB. Before memory stick and a wider usage of CD recorders, it was used as data carrier. 'Modern memory sticks have a memory capacity measured in GB while floppy disks only have memory capacity of 1.44MB, indicating that floppy disks are becoming obsolete.

General Principles of Teaching Aids

Principle of Selection

- They should suit the age level, grade level and other characteristics of the learners.

- They should have educational value besides being interesting and motivating.
- They should be the true representation of the real things.
- They should help in the realization of desired learning objectives.

Principle of Preparation

- As far as possible, locally available materials should be used in the preparation of an aid.
- The teachers should receive some training in the preparation of aids.
- The teachers themselves should prepare some of the aids.
- Students may be associated in the preparation of aids.

Principle of Physical Control

- This principle relates to the arrangement of keeping aids safely and also to facilitate their lending to the teachers for use.

Principle of proper Presentation

- Teachers should carefully visualize the use of teaching aids before actual presentation.
- Adequate care should be taken to handle an aid in such a way as no damage is done to it.
- The aid should be displayed properly so that all the students are able to see it.
- As far as possible, distraction of all kind should be eliminated so that full attention may be paid to the aid.

Sound Card

Soundcard is a device used for sound creation and production by means of computer speakers.

CD-ROM and DVD Drive

- CD-ROM drive is used for reading CD media.
- DVD drive is used for reading DVD discs. DVD disc capacity ranges from 4.7 to 18GB.

Graphic Cards

Graphics card is responsible for image processing and displaying it on a monitor. It has its own graphics processor and memory. Image quality depends on the strength of these components.

Modem and Connectors

- Modem enables computers to communicate via telephone lines. They connect computers to the Internet.
- Connectors or ports are slots visible in the back and the front side of a computer.

Universal Serial Bus (USB)

Universal Serial Bus (USB) is used to connect various devices (mouse, keyboard, USB memory).

Serial Port and Parallel Port

- Serial port is used for example in connecting a mouse (labeled COM1 or COM2).
- Parallel port is used for connecting a local printer (LPT1 or LPT2).

Network Port and Firmware

- Network port is used for connecting computers to a network.
- Firmware is used for connecting computers and audio-video devices (digital cameras, etc.).

Measurement Units

Bit (binary digit) is the basic unit used to measure the amount of information. A byte or octet contains eight bits.

- 1 KB (kilobyte)- 1024 B (approx. 1000 B)
- 1 MB (megabyte)- 1024 KB (approx. 1000 KB)
- 1 GB (gigabyte) -- 1024 MB (approx. 1000 MB)
- 1 TB (terabyte) – 1024 GB (approx. 1000 GB)

Blu-ray Disc

Blu-ray disc (BD)- the successor to DVD, is an optical disk storage, it comes in different capacities, depending on how many layers it has and the capacity of each layer. Currently, the capacity of one layer is between 27 GB and 33 GB, while the overall capacity is the product of the number of layers and capacity of each layer.

Memory Card and USB Stick

Memory card is a type of flash memory used to store data in digital cameras, cell phones, MP3 players etc. USB Stick is a data storage device. It features small dimensions, relatively high

capacity, reliability and speed. It belongs to the type of flash memory that remembers data, even when not under voltage i.e. they do not need electric power to maintain data integrity.

Limitations of ICT

- ICT supposedly endangers the child's social development.
- ICT is supposedly to endanger the child's intellectual development.
- Decreasing of child's emotional development
- Child's health is endangered.
- Misuse of ICT A
- Complete control over the child's usage of ICT is today
- Practically impossible
- Expensive

Internal and External Hard Drive

There is a difference between an internal hard disk drive, which is embedded in the computer case, and an external hard disk drive, which is connected to a computer by using an appropriate cable or USB port, and is usually used to transfer data from one computer to another or for backup.

Mouse

Mouse is an input device that facilitates work with the graphical user interface (GUI).

The mouse transmits hand movements and the screen displays the cursor (mouse pointer) movements. They are divided into mechanical and optical (with respect to a transfer movement), and wired and wireless (with respect to connection).

Track Ball and Keyboard

- Trackball, unlike a mouse, is not movable. Hand movements are transmitted to the screen by rolling the ball which is located on the upper side of the device.
- Keyboard is used for data entry and issuing commands. They can also be wired or wireless.

Scanner

Scanner is used to load data (image, text, etc.) from the printed material into a computer.

The result of scanning is an image, but with special programs, if we scan the text, we can get a

text as a result. Software used to recognize text from image is called a text recognition tool.

Touch Pad and Light Pen

- Touchpad is used for transmission of hand movement, but unlike working with a mouse, the user is the one who determines the position of the cursor by touching the touchpad.
- Light pen enables handwriting on screen and can be used as a mouse. It requires an appropriate monitor type.

Joystick

Joystick: mainly used in computer games. Unlike a mouse, it has many buttons which allow control over game objects.

Microphone

Microphone is a device that converts sound into an electrical signal, which can be stored on a computer. It is mainly used for recording sound, communication between players in online games, in combination with a web camera in video conferencing, for converting voice into text on a computer (speech-to-text processing (e.g., textual files or emails), etc.

Structure of Monitor

Monitor displays images from the computer, it enables us to see, work and control computers. In other words, working on a computer without a monitor would be inconceivable. Common types of monitors, with regard to manufacturing technology, are the CRT and LCD. CRT monitors have been present on the market for a long time, and other technologies are pushing them out. They are based on cathode tube technology. LCD monitors use liquid crystal technology. In comparison with CRT monitors, LCD monitors use less electrical energy, do not emit radiation and their price is higher, however due to smaller dimensions, more attractive design and a good picture quality, they are pushing CRT monitors out of the market. Monitor size is expressed by the size of screen diagonal and measured in inches ("). Picture quality is expressed with the notion of resolution, which is a number of horizontal and vertical dots (pixels) (e.g. 1920x1080).

Printer

Printer is a device used for printing data from a computer onto a paper. We distinguish between local printer (connected directly to the computer) and network printer (connected directly to network using a network card). Also, printers also differ according to print

technology: dot matrix, laser, inkjet, thermal printer and plotter.

Dot Printer and Laser Printer

Dot matrix printers are the oldest, with the lowest price of print per paper, they are slow, make a lot of noise while printing, and are mostly appropriate for printing text.

Laser printers are similar to photocopy devices when it comes to technology. They have exceptional print quality, speed and are quiet. Downsides of laser printers are their high price and high price of toners.

Merits of HTML

- HTML is simple but powerful formatting language to use.
- The simplicity allows anyone to create web pages.
- HTML documents are device independent (works on any platform)
- The web pages can be linked together using links
- The pages can be updated at any time.

Demerits of HTML

HTML does not have any programming capabilities.

- Require expertise knowledge.
- Time consuming.
- Difficult to remember coding

Projector

Projector is a device used to project a computer image or other images from independent devices, such as DVD players, Blu-ray player, etc. onto canvas or a wall.

Inkjet Printers and Thermal Printer

- Inkjet printers have a high print quality (somewhat lower in comparison with laser printer), they are quiet while printing, and have low initial investment. Ink price, especially color ink, can cost as much as the printer itself. Printing technology is based on

ink dispersion from container onto paper.

- Thermal printer, as its name states, leaves a print on the paper by utilizing heat. They use paper sensitive to heat, feature small dimensions; they are quiet while printing and relatively cheap. They are usually used for printing receipts, and owing to that they are called POS printer (printer of sale). Also, they are used as calculator printers and due to their small dimensions, as portable printers.

Plotter

Plotter is used for printing large drawings (up to A0). They are extremely expensive and used only for professional purposes, such as in designing firms for printing technical drawings (blueprints).

Storage Devices and Touch Screen

- Storage devices, due to necessity for writing and reading data, they are classified as input/output devices.
- Touch screen (i.e. monitor sensitive to touch) is out device while displaying computer image, and at the same time input device while receiving manual orders.

Software

Software is, unlike hardware, intangible part of the computer. It consists of a sequence of commands, written according to strict rules. Programs are written by programmers, in various programming languages.

Operating System

Operating system is a program which manages computer hardware. First computers did not have operating systems; they had programs that were directly loaded into the computer (e.g. punch cards). Today, computers have an operating system which loads into the computer's memory during its startup. Computer functions are based on its operating system. Within operating system, drivers (responsible for the functioning of a computer) and various utility programs (responsible for the functionality of a computer) are installed. The most famous operating systems are:

1. Linux (Debian, Ubuntu, Fedora, Knoppix) open source software
2. Microsoft Windows (XP, Vista, 7) proprietary software

3. Mac OS X (Cheetah, Panther, Snow Leopard) proprietary software

Application Software

Application Software (Utility programs) are all programs that users use to perform different tasks or for problem solving. Users, according to his/her needs, install the appropriate utility software. Computer functions and tasks that computers can perform are defined by the installed utility software. Utility software can often cost more than computer hardware unless the software is open source.

Text Processing Software and Spread Sheet Calculation Software

- Text processing software is used for creating and forming text documents and nowadays, they can contain images, charts and tables. Examples of such programs are OpenOffice.org Writer (open source software) and Microsoft Word (proprietary software).
- Spreadsheet calculations software is used for performing various calculations and presentation of results in charts. Examples of such programs are Open Office, Writer (open source software) and Microsoft Excel (proprietary software).

Software for Presentation

Software for presentations is used to create professional presentations that consist of slides with graphical and textual elements. Such a presentation can afterwards be displayed as a "slide show" by using a projector. Examples of such programs are OpenOffice.org Impress (open source software) and Microsoft PowerPoint (proprietary software).

Magnifier and On Screen Keyboard

- Magnifier is used to enhance a part of the screen.
- In on Screen Keyboard text is entered using a mouse to click on the on-screen keyboard.

Importance of Audio Visual Aids

- They are best motivators
- Reduce verbalism
- Gives clear images
- Vicarious experience
- Provide variety in methods of teaching and learning

- Freedom to child
- Opportunities to handle and manipulate
- Contributes to increased retentively
- Based on maxims of teaching
- Helpful in attracting attention
- Helpful in fixing up new learning
- Saving energy and time
- Gives reality and vividness to learning situations
- Meeting individual differences
- Encouragement to healthy classroom interaction
- Spread of education on a mass scale

Narrator

Narrator is commonly used by users with visual impairment - it can read text displayed on monitor, it tells current cursor position, and describes certain events, like warning and error messages generated by OS.

Function of Windows Speed Recognition

Windows Speech Recognition enables speech recognition, i.e. recognizes spoken word, transfers it to text and enters it into a document; therefore it enables you to dictate a text to a computer, to browse the web using your voice etc.

Computer Network

Computer network is comprised of at least two, connected, by wire or wireless, computers that can exchange data i.e. communicate. There are many reasons for connecting computers into a network, and some of them are:

Exchange of data between users that have network access,

- Access to shared devices, such as network printers, network disks, etc.,
- Enables user communication and socializing, etc.

LAN (Local Area Network), WLAN (Wireless Local Area Network and WAN (Wide Area Network)

LAN (Local Area Network)

It is defined as, a network that covers a relatively small geographical area- it connects computers within a firm or household by wire.

WLAN (Wireless Local Area Network)

It is defined as, a network that covers a relatively small geographical area - it connects computers within a firm or household wirelessly.

WAN (Wide Area Network)

It is defined as, a network that covers a relatively large geographical area - it connects a greater number of computers and local networks.

Client and Server

Relationship client - server is defined in the following manner: client sends requests and server responds to those requests. We can use Internet as the best known example. User's computer, connected to the Internet, sends requests to a certain web page (by entering page address into the Internet browser Address bar), and the server responds. Web page is loaded into the user's computer Internet browser as a result of server response. From this example, we can see that communication between client and server depends on connection speed (bandwidth).

Internet, Intranet and Extranet

- Internet (network of all networks) is a global system comprised of interconnected computers and computer networks, which communicate by means of using TCP/IP protocols. Although, in its beginnings, it emerged from the need for simple data exchange, today it affects all domains of society.
- Intranet is a private network of an organization to which only authorized employees have access (login and password).
- Extranet is part of Intranet, to which independent collaborators have access.

Different Way to Connect the Internet

- Dial-up Internet access method uses a modem (56k) and a telephone line
- Broadband is characterized by a high speed data transfer, permanent access to the Internet, and thus the risk of unauthorized access to the network or your personal computer.

Different Connection Methods of Internet

Mobile

Internet is connected using a mobile network (GPRS, EDGE, UMTS, HSPA)

Satellite

These are commonly used in parts of the world where there is no proper infrastructure and there is no other way of accessing the Internet

Wireless (Wi-Fi)

Data is transferred between computers by using radio frequencies (2.4 GHz) and the corresponding antennas.

Cable Internet

It is used in connecting to the Internet through television cable network using a cable modem

Broadband

It is characterized by a high-speed data transfer, permanent access to the Internet, and thus the risk of unauthorized access to the network or your personal computer. In the beginning of broadband Internet access, due to underdeveloped communication infrastructure, Internet providers charged based on the data traffic but not time spent on the Internet (unlike dial-up Internet access). Today, in large cities, telecommunications infrastructure is developed, therefore Internet providers do not charge money based on the time spent on the Internet or the amount of transferred data but they do charge by access speed.

E-Commerce and E-Banking

- E-commerce is a form of trade that allows customers to browse and purchase products online.
- E-banking allows users to have control over their accounts (view balances and transactions), transactions from one account to another, credit payment, shopping vouchers for mobile phones etc. The benefits are saving time (which would otherwise be spent waiting in lines), lower service fees, and access from anywhere, anytime.

E-Government

In E-Government, information technology is used to provide better public access to government information, therefore providing citizens with their human right to information.

Different Classification of Projected Aids

Projected Aids

- A projected aid is one which items to be observed is projected on a screen using electronic or Mechanical device.
- Very effective because movements can be added to sound and hence a realistic experience can be provided.

Magic Lantern

- It is used for projecting pictures from a transparency on a wall or screen.
- When the figure or illustration is very small and it is required to be shown to the entire class, A transparent slide of the small figure is prepared.
- Then, this slide is placed into the slide carrier part of the magic lantern.
- This magic lantern device projects it on the screen by enlarging its dimension and making the vision more clear and sharp.

Film & Film Projector

- Films are in the form of motion pictures.
- Films enrich learning by presenting a series of meaningful experience involving motion.
- Can enlarge or reduce the actual size of the object to suit the need.
- Film strips & Film strip Projector.
- A film strip is piece of nonflammable safely film, 35 mm wide.
- Length may vary according to requirement and can be up to about one meter.

One film strip may contain 10 to 50 pictures frames.

- The pictures in film strip may constitute a connected series of drawings, photographs, diagrams or combination of these and illustrated a single sequence.
- The pictures may in color or in black or white.
- Some films strips are accompanied by commentary recorded separately, such film strips is called sound film strip.

Slides & Slide Projector

- The pictures or diagrams which are drawn in suitable medium is called slide. (Glass or Transparency)

- Slide can be enlarged by projecting it on a screen with a slide projector.
- Slide projector is an instrument equipped with a powerful light source and a carrier for holding slides of suitable size.
- Some slide projectors use drums in which many slides can be loaded in proper sequence in advance.
- It is also possible to record the narration in a tape recorder that could be hooked up to the projector in such a way as to give the necessary commentary without the help of the teacher.

Episcope & Epidia scope

- Episcope is used for the projection of opaque objects.
- Pictures, photographs, drawings or an opaque material within the size of the platform of the episcope can be projected using Episcope.

Episcope used to enlarge book diagrams and illustrate complex pictures.

During Epi-projection the class room should be totally darkened in order to get a clear and bright image.

- Epidiascope can project opaque as well as transparent objects.

OHP

- This is called Over Head Projector because it projects the image behind and over the head of the teacher.
- The transparent visual is placed on a horizontal platform at the top of the light source.
- The teacher can place the transparent plate on the platform and write or draw on it anything that is to be visualized using suitable pen.

Television

- TV is very exciting and efficient means of mass communication.
- Educational Television programmes aim at education rather than entertainment.
- One teacher or specialist can render teaching session and pass information to millions of viewers all over the world.

LCD Projector

- LCD (Liquid Crystal Display) is the most advanced and sophisticated projected aid.

- In the brighter area of educational technology, we can replace all other projected and non-projected aids with only a single LCD projector and computer system.
- It can present a topic by using computer made slide, graphs, pictures, video clips, movies and special effect.

DLP Projector (Digital Light processing Projector)

- Most advanced and sophisticated projected aid.
- Brighter image
- Lightest and smallest

E-Learning

E-learning consists of all forms of learning and/or knowledge transfer that are based on electronic technologies. This term will mostly be used to describe learning and/or dissemination of knowledge without direct teacher-student contact, while using ICT technology.

Advantages of E-Learning

Advantages of e-learning greatly coincide with learning from books, with a difference that the books come only in text and graphic form on paper, while e-learning materials have multimedia features, and the ability to remotely monitor lectures in real time. To illustrate, physicians can observe medical procedures that are currently taking place in another hospital; students can attend classes that are currently taking place at another university.

Some of the advantages for students:

- Adaptive time, place and pace of learning
- Access and availability of education
- Unlimited repetition of lectures
- Reduced cost of education
- Multimedia environment (video, audio, text)

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Benefits of E-Learning for Teachers and Institutions

Some of the benefits for teachers and educational institutions:

- Reduced costs of conducting teaching (reduced travel costs and accommodation, reduced or eliminated costs for classrooms, etc.)
- Much larger education user base
- Once created, e-education allows educational staff to invest a much shorter time commitment to implement the respective education
- Simple and easy upgrade of materials, etc.

Advantages of Teleworking

- Reduced utility costs and employees transportation reimbursement
- Reduced time for traveling to and from work
- Adapting working conditions to personal habits flexibility
- Possibility of working in the convenience of a person's home (or other places that are not an office)
- Lower stress level, provided there is a good organization and coordination , i.e. the flow of information between employees is unconstrained
- An excellent solution for people with disabilities or long term illness, parents, or those who enjoy traveling

E-Mail

E-mail is the transmission of text messages via Internet. Using attachment, users can exchange files like images, multimedia, documents, etc. E-mail addresses consist of a username, @ sign ("at") and a domain name, e.g. mark.marich@ITdesk.info, where "mark.marich" is the username and "ITdesk.info" is the domain name.

Disadvantage of Teleworking

- Physical isolation and lack of direct social contact with colleagues
- Not suitable for everyone - requires a certain dose of self-control and motivation

- Less opportunity for advancement due to lack of traditional vertical hierarchy in the company, however horizontal hierarchy is created, where no one is above anyone, but everyone has their own duties and obligations

IM (Instant Messenger)

IM (Instant Messaging) service is used for instant message exchange and real time communication. Besides textual communication and video conferencing, they allow file transfer as well. Some of the popular IM services are: Windows Live Messenger, Skype, and Google Talk.

VoIP (Voice over Internet Protocol)

VoIP (Voice over Internet Protocol) is a term for technology that enables digitalization and transfer of sound (and multimedia sessions) over the Internet. Therefore, it enables sound communication much like the telephone call. In short, VoIP protocol enables audio/video communication between users.

RSS (Really Simple Syndication)

RSS (Really Simple Syndication) is a set of web formats, used by web pages that are being constantly updated, considering that by using RSS, new information, articles, etc. can be published. RSS enables us to, with use of appropriate software, read new titles and access news summaries. Users can, with appropriate software, obtain news from various portals and receive information without the need to visit all of them independently, while publishers can distribute news easily to end-users who are monitoring their RSS feeds/channel.

Blog (Web Log)

Blog (web log) is a form of Internet media in which users can re-post or write their opinion on any topic. It can be personal (a form of Internet diary), thematic, or in the form of journals. Blogs are popular because end-users do not need to have advanced computer knowledge, or knowledge of webpage creation, in order to be present on the Internet and write about various topics, i.e. state their opinion. Blogging has greatly facilitated the exchange of interests, ideas, opinions, and contributed to realization of human right to free expression. Some use blogs to share their art, personal aesthetic, or to collect virtual clippings from websites online.

Advantages of Projected Aids

- Teacher can prepare material in advance.
- Saves time in classroom.
- Provides greater enjoyment in learning
- Stimulates more effective learning
- Increases retention
- Increases attention
- Enlarges or reduces actual size of objects.
- Can provide multi-sensory/multimedia approach.
- Brings distant past and the present into the classroom.
- Can make class/presentation more effective.

Posts

Entries published on blogs are called posts, which may allow a space below for visitors to leave their comments. Besides text, it is possible to exchange multimedia files, such as photos or videos. In order to share your texts, pictures, video clips, etc., it is necessary to create a personal account on websites that offer this type of service. In case you simply wish to browse the content, it is not necessary to open a personal account.

Podcast

Podcast (POD - Personal on Demand+ Broadcast) is a digital file containing audio or video content. It is distributed over the Internet using RSS technology.

Virtual Community

The term "virtual community" encompasses a group of people who communicate via social networks, forums, IM service, blogs. We call it a community, since they are a collection of people who participate in a group of their interests, thus forming a virtual community, because it does not involve physical contact, but is carried out by means of ICT technologies.

Social Networks and Forum

Social Networks

These are free online services that provide users with various forms of presentation and communication. Some of the more popular social networks are: MySpace, Facebook, Twitter, and Google+.

Forum

It is defined as a service that allows users to exchange opinions. It can be compared to bulletin boards on which participants leave their messages, while others leave their comments. Examples of forums are: Reddit, TED Conversations.

MMO (Massive Multiplayer Online)

MMO (Massive Multiplayer Online) are network games, played by a large number of players via the Internet. One of the sub-genres of MMO is a MMORPG (Massive Multiplayer Online Role-Playing Game), and its most popular representative is a game called World of Warcraft.

Chat Room, Video Sharing and Photo Sharing Website

- Chat room is a webpage or part of a webpage which allows a real-time information exchange. Unlike the instant messaging writing software, it does not require computer installation, only a web browser and Internet connection.
- The most popular service for video clip sharing is YouTube.
- Picture or photograph exchange services are Photo bucket, Flickr, Instagram etc.

Computer Virus

Virus is a malicious program with the possibility of self-replication. In a computer it seeks other files, trying to infect them and the ultimate goal is to spread to other computers. Depending on what malicious code it contains, it may cause more or less damage to your computer.

Trojan Horses

Trojan horses can be presented in the form of games or files in e-mail attachments, with different purposes, such as: to allow remote control to your computer, delete data, allow spreading of viruses, send spam from the infected computer, etc. Unlike viruses and worms, it

does not replicate and it needs to be executed, that is, it is not autonomous.

Worms

Worms like viruses, they possess the ability of self-replication, and however they cannot infect other programs. Their usual intent is to take control of your computer and allow remote control by opening, the so-called, and 'backdoor'.

Spyware and Adware

- Spyware is a program that secretly collects data about computer users (passwords, credit card numbers, etc.), therefore it spies on users. The most common spyware infections usually occur while visiting suspicious sites with illegal or pornographic content.
- Adware is a program that uses collected data about a user in order to display advertisements in a way regarded as intrusive. It usually manifests itself in the form of many pop-up windows that display ads.

Disadvantages of Projected Aids

- Expensive.
- Teacher must know to use.(Technical knowledge)
- Requires electricity and other infrastructure.
- Requires a darkroom.
- Requires smart room facilities.
- Chance for diverting from study.
- Require maintenance.
- Require techno friendly (ICT enabled) teaching materials.

Prepared by:-

Muhammad Imran

EDU430 ICT in Education (11/09/2020) 5:00pm

20 MCQ'S easy thay.

2 Marks Questions.

21-Define compression.

22- Define information technology.

23- what is E-mail.

24- What is ISO?

25- Define BIOS.

3 marks Questions

26- Differentiate between teaching, training and instructions.

27- what is bps?

28- what is IP address?

29- Define hacker?

30- Define filter.

Long Question 5 marks

31- what do you know about Technology integration. Discuss in detail.

32- How mathematical mindset can use technology and pedagogy for teaching purpose.

33- Use of ICTs for professional development of teachers

34- effect of emotional, psychological and cultural on use of ICTs

35- how ICTs help illiterate youth.

edu430 Past Paper Shared by 2019

Define information technology

Download

ISO

Internet

Pixels

Host

Firewall

Uses of charts in classroom

Forgot rest...

Long ques were from past paper the file is uploaded yest in group files..

edu430 Past Paper Shared by 2019

10 MCQs, mostly from handouts quotations.

Q1. Define Folder?

Q2. Define ISO?

Q3. Define ERIC?

Q4. Define ICT?

Q5. Define associate?

Q6. Define internet?

Q7. Describe the benefits of ICT for the teachers and the institutions?

Q8. benefits of ICTs in school and out of school

Q9. Briefly define why ICT benefit for illiterate youth?

Q10. How can educational content for dissemination through ICTs

Define BIOS?

What is attachment?
define Inline image?
Define Drum scanner?
What is Dotted decimal notation? marks
What is external viewer?
Difference b/w virtual reality and virtual world?3 marks
Differential impact of ICT on subgroups of boys and girls?3 marks
How private-public sector partnership in introducing ICT in education can cut down the cost of learning??5 marks
Describe How Handheld devices effective in education (especially for the teacher professional development and administration)??5 marks

• mcqz saray bakwas....40.... saray babay ke kahanio say..... no idea about that... like
galib kahan peda hua..?
dr. ashraf k aba g kis zuban k mahir thay?
howard university k kitnay students pay test pass kia first time mai?
books names
quotes by authors
muslim school have which level?
in 1969 how many universities in Pakistan?
book authors names bohat kuch jo k
takreeban 8,9 ko chor k baki sub babay ke kahanio mai say he thay....
Subjective.....

1. Define GCG? 2
2. Define Icon? 2
3. what is the meaning of backbone in computer? 2
4. define Dichroic mirror? 2

5. what is computer search service? 2
6. define densitometer? 3
7. define resampling? 3
8. what is FDDI? 3
9. define header? 3
10. What is the meaning of computer program? 3
11. How to type of learning strategies be selected by use of ICT upon special need and disadvantaged students? What is the gender in terms

difference in this regard? 5

12. How handheld devices which are enabled 3g to use in support of ICT especially educational professional development and management for teachers? Which are the best practices are used in this regard? 5

13. How ICT components in education support by donors be quantified and identified? 5

Very best of luck for all.....

edu430 final term paper

edu430 final term papers

edu430 Past Paper Shared by 2019

Define hard disk drive.(2)

Define GB(2)

Define DOS(2)

Define cookies(2)

Define Crush(3)

Define Internet(3)

Define firewall(3)

Define IP(2)

Define IP datagram(3)

What is the meaning of bookmark?(5)

What are the Advantages of Projected aids?(5)

Ict in edcation fields(5)

Instant feedback and summative assessment(5)

EDU430 Final Term Paper

10 MCQs, mostly from handouts quotations.

Q1. Define Folder?

Q2. Define ISO?
 Q3. Define ERIC?
 Q4. Define ICT?
 Q5. Define associate?
 Q6. Define internet?
 Q7. Describe the benefits of ICT for the teachers and the institutions?
 Q8. benefits of ICTs in school and out of school
 Q9. Briefly define why ICT benefit for illiterate youth?
 Q10. How can educational content for dissemination through ICTs
 Define BIOS?
 What is attachment?
 define Inline image?
 Define Drum scanner?
 What is Dotted decimal notation? marks
 What is external viewer?
 Difference b/w virtual reality and virtual world?3 marks
 Differential impact of ICT on subgroups of boys and girls?3 marks
 How private-public sector partnership in introducing ICT in education can cut down the cost of learning??5 marks
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